

Quiz

A fuse must be selected for certain application. You may choose from fuses rated to “blow”, when the current exceeds 1.5 A, 3A, 4.5 A or 5 A. If the supply voltage is 110V and the maximum allowable power dissipation is 500 W, which fuse should be chosen, and why?

Austin Peay State University

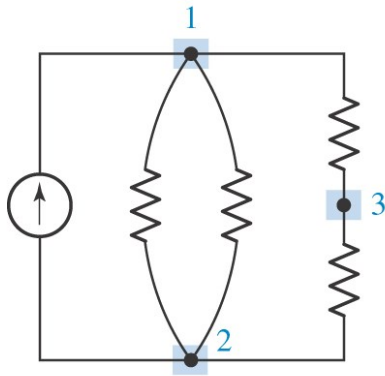
PHYS 3050: INTRO to CIRCUITS

Voltage and Current Laws

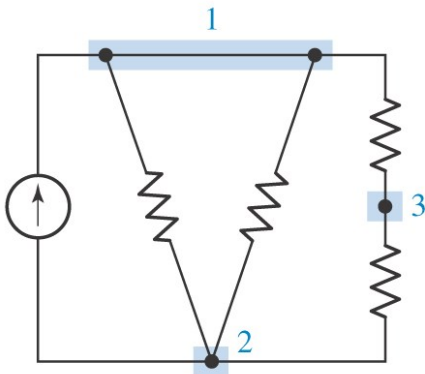
Nodes, Paths, Loops & Branches

Network, which contains a number of simple elements and a set of connecting leads (wires with zero resistance) is called **lumped-parameter network**

Network, which contains an essentially infinite number of vanishingly small elements is called **distributed-parameter network**



(a)



(b)

Node – point at which 2 or more elements have common connection

Path – the set of nodes and elements that we have passed through

Loop – closed path

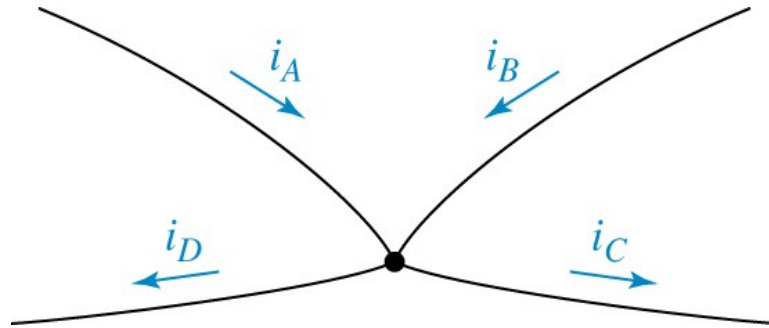
Branch – single path in a network, compose of one simple element and the node at each end of that element

(a) A circuit containing three nodes and five branches.

(b) Node 1 is redrawn to look like two nodes; it is still one node.

Kirchhoff's Current Law (KCL)

The algebraic sum of the currents entering any node is zero



Consider the node shown in Fig. 3.2. The algebraic sum of the four currents entering the node must be zero:

$$i_A + i_B + (-i_C) + (-i_D) = 0$$

It is evident that the law could be equally well applied to the algebraic sum of the currents *leaving* the node:

$$(-i_A) + (-i_B) + i_C + i_D = 0$$

We might also wish to equate the sum of the currents having reference arrows directed into the node to the sum of those directed out of the node:

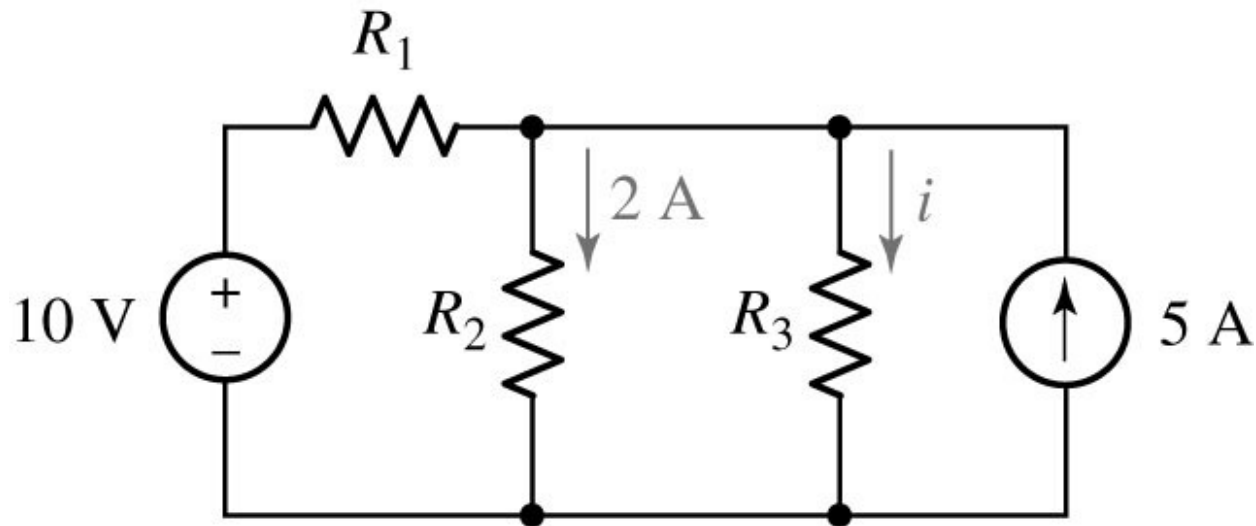
$$i_A + i_B = i_C + i_D$$

which simply states that the sum of the currents going in must equal the sum of the currents going out.

Kirchhoff's Current Law (KCL)

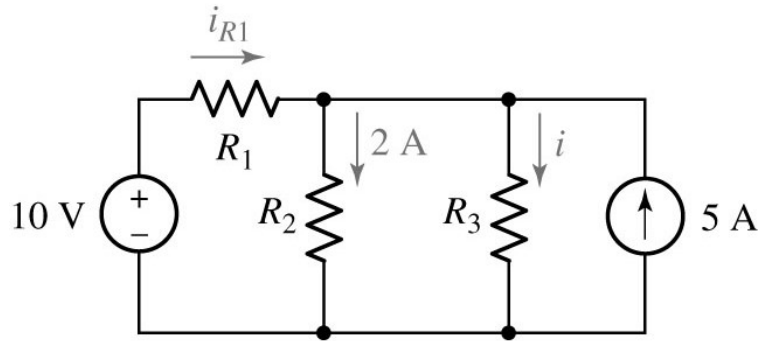
Example

For the circuit in the Fig. below compute the current through resistor R_3 if it is known that the voltage source delivers a current of 3 A.



Kirchhoff's Current Law (KCL)

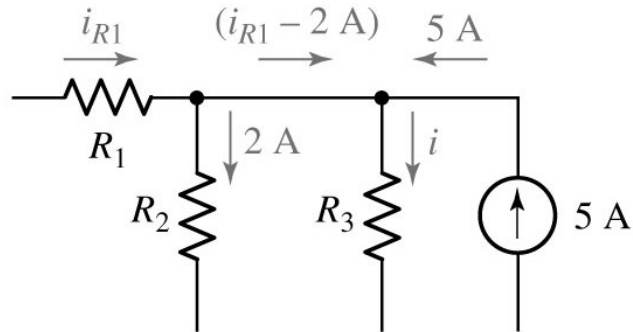
Example



(b)

$$i_{R1} - 2 - i + 5 = 0$$

$$i_{R1} = 3 \text{ A}$$

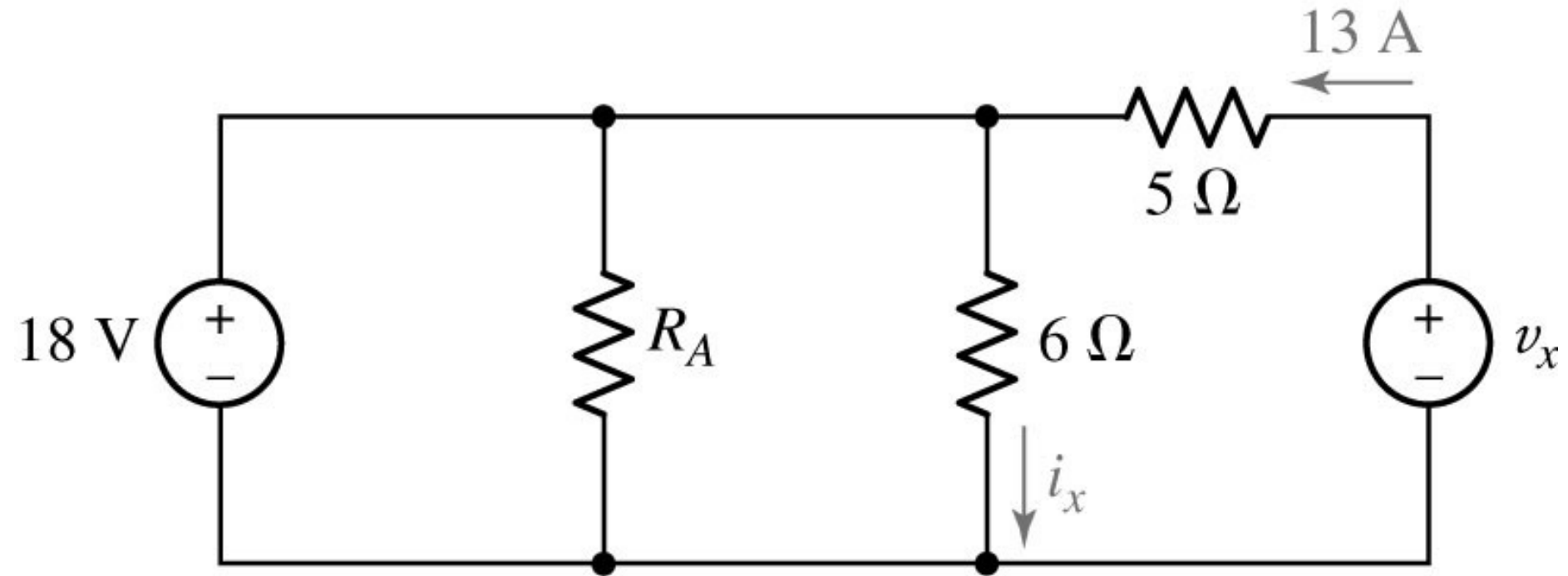


(c)

$$i = 3 - 2 + 5 = 6 \text{ A}$$

Kirchhoff's Current Law (KCL)

Practice

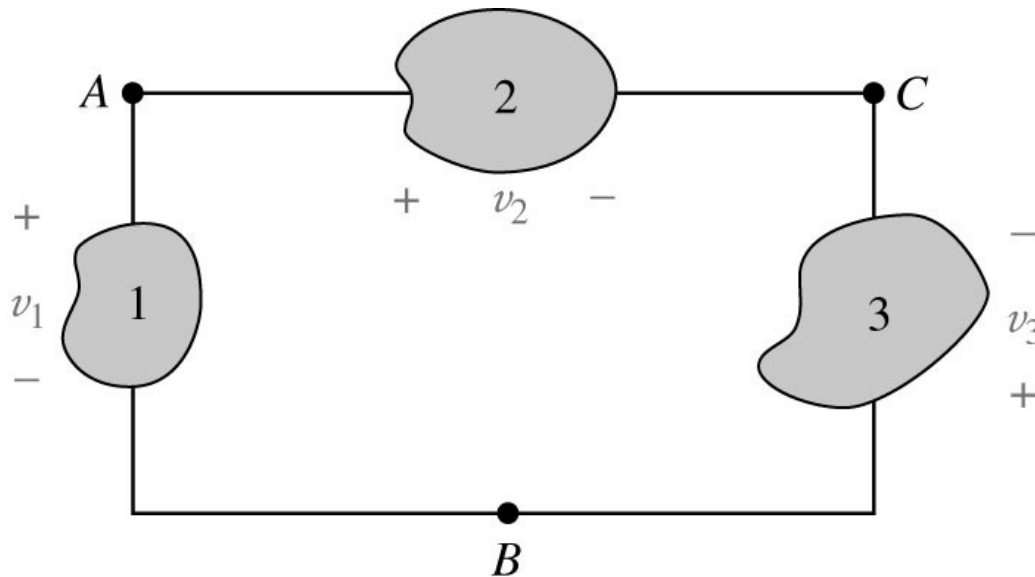


Count the number of branches and nodes in the circuit in the Fig. If $i_x = 3\text{ A}$ and 18 V source delivers 8 A of current, what is the value of R_A ?

Kirchhoff's Voltage Law (KVL)

Voltage is independent of path chosen to get from A to B

The algebraic sum of the voltages around any closed path is zero



$$v_1 = v_2 - v_3$$

$$-v_1 + v_2 - v_3 = 0$$

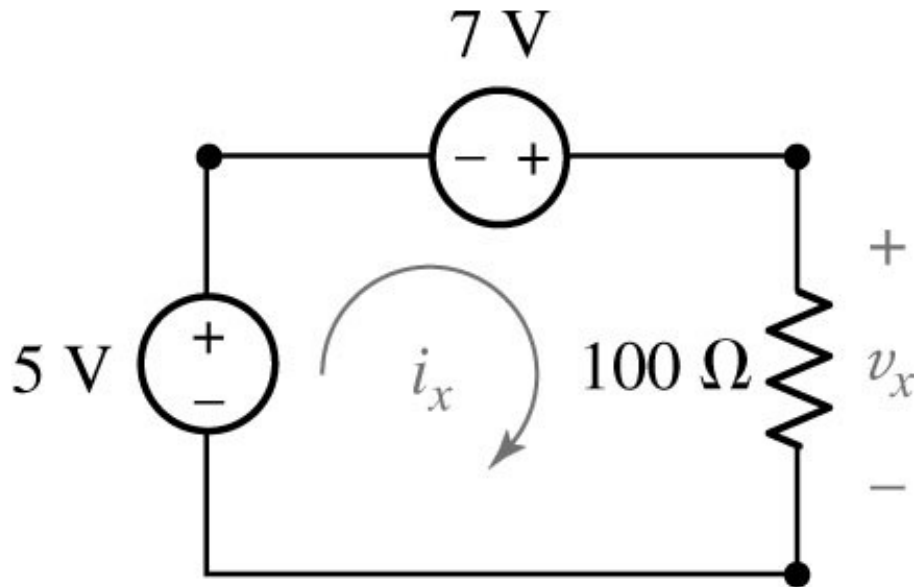
$$\sum_{n=1}^N v_n = 0$$

Usual way to apply KVL: Moving around closed path in a clockwise direction and writing down directly the voltage of each element, whose (+) terminal is entered, and writing down the negative of every voltage first met at the (−) sign.

Kirchhoff's Voltage Law (KVL)

Example

In the circuit find v_x and i_x



$$-5 - 7 + v_x = 0$$

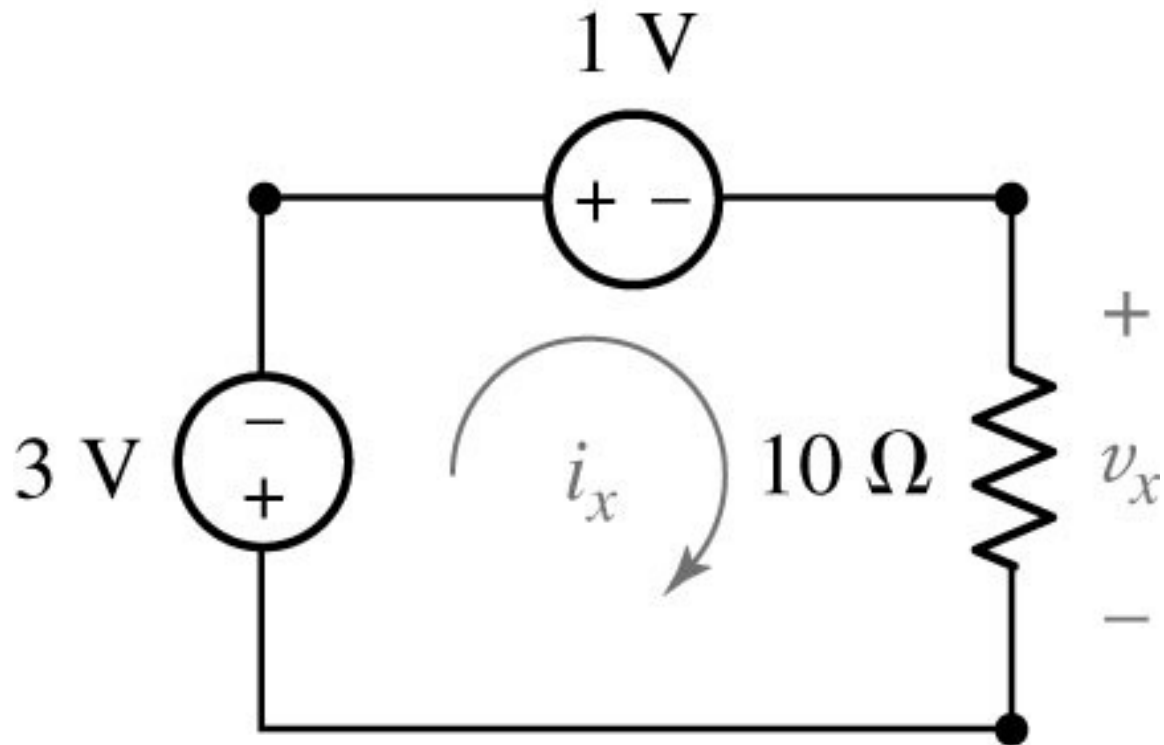
$$v_x = 12 \text{ V}$$

$$i_x = v_x / 100 = 120 \text{ mA}$$

Kirchhoff's Voltage Law (KVL)

Practice

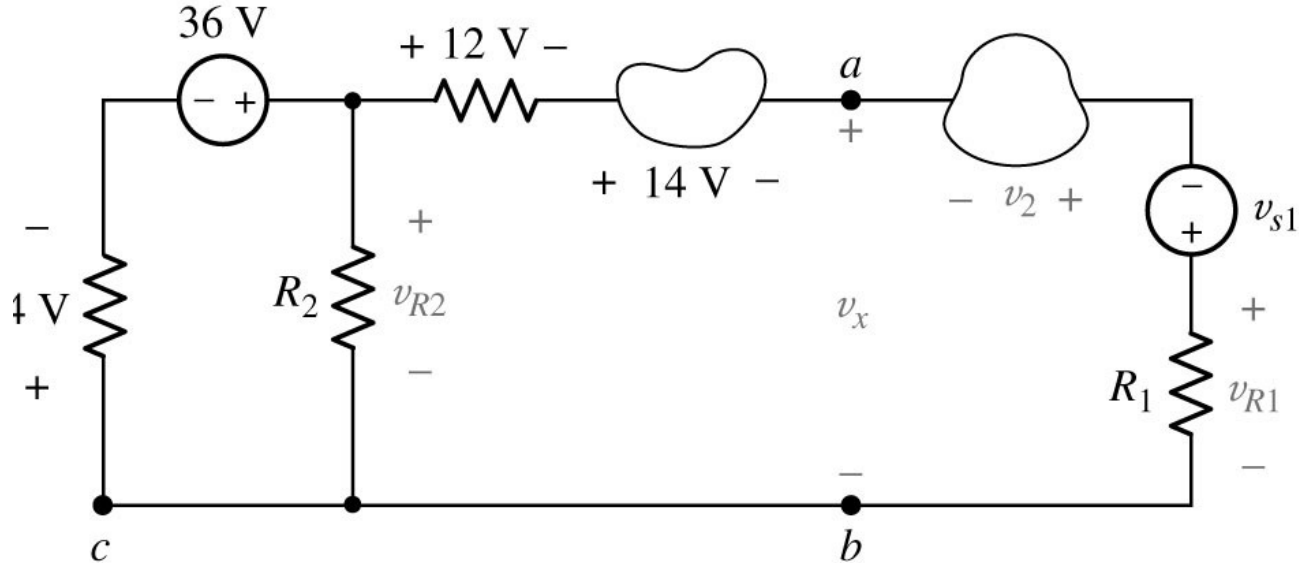
In the circuit find v_x and i_x



Kirchhoff's Voltage Law (KVL)

Example

In the circuit there are eight circuit elements; voltages with plus-minus pairs are shown across each element. Find v_{R2} (the voltage across R_2) and the voltage labeled v_x .



$$4 - 36 + v_{R2} = 0$$

$$4 - 36 + 12 + 14 + v_x = 0$$

$$v_{R2} = 32 \text{ V}$$

$$v_x = 6 \text{ V}$$

Kirchhoff's Laws (KCL + KVL)

The **key** to correctly analyzing the circuit is to first **methodically label all voltages and currents** on the diagram.

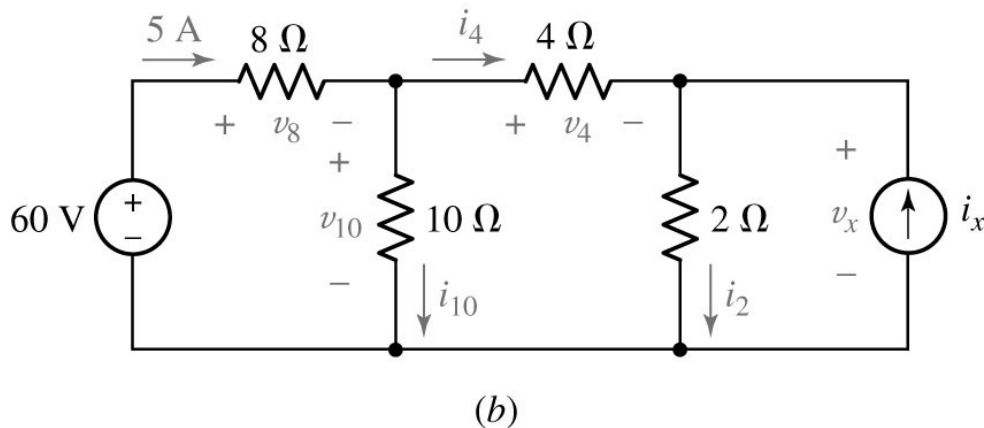
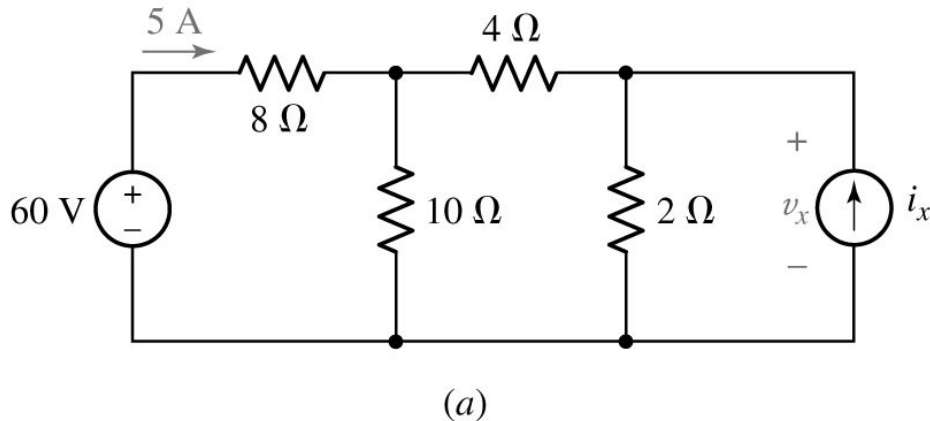
Next step: carefully write KCL or KVL equations

Next step: apply Ohm's law if necessary

Kirchhoff's Laws (KCL + KVL)

Example

Determine v_x in the circuit



$$\text{KCL: } i_2 = i_4 + i_x$$

$$\text{KVL: } -60 + v_8 + v_{10} = 0$$

$$\text{KVL: } -v_{10} + v_4 + v_x = 0$$

$$v_8 = 40 \text{ V}$$

$$v_{10} = 60 - 40 = 20 \text{ V}$$

$$v_x = 20 - v_4$$

$$i_4 = 5 - i_{10} = 5 - v_{10}/10$$

$$i_4 = 5 - 20/10 = 3 \text{ A}$$

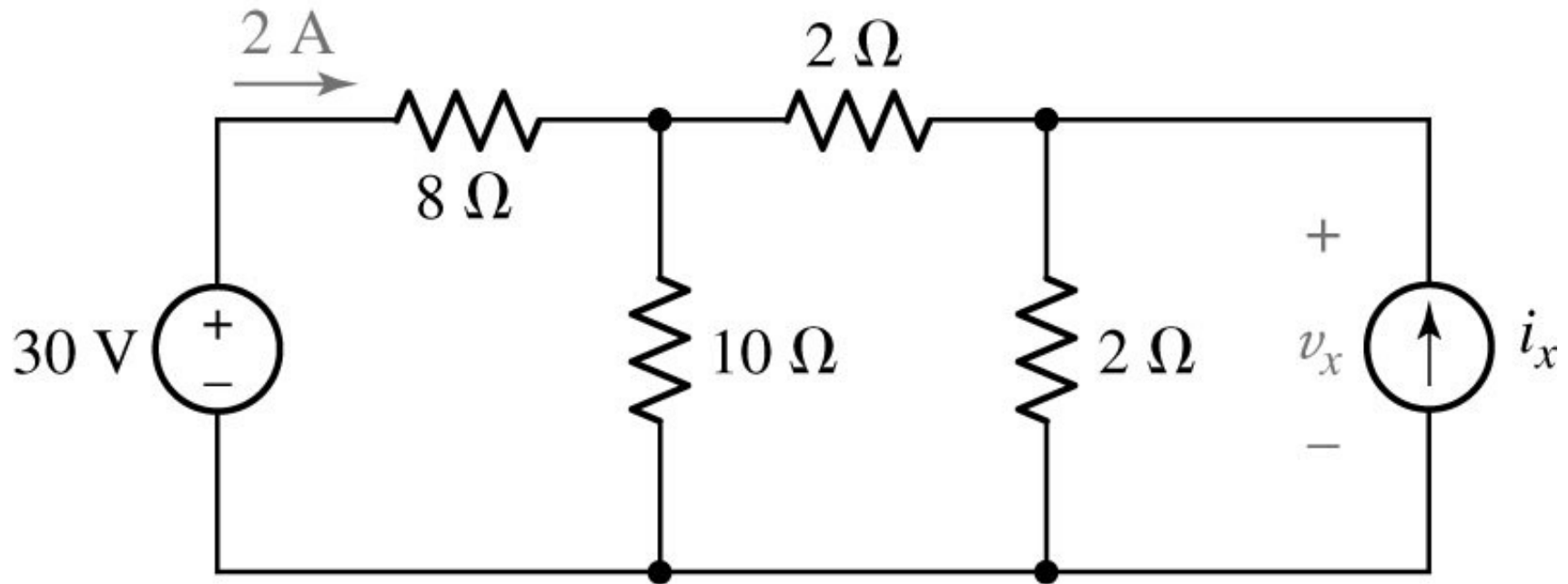
$$v_4 = (4)(3) = 12 \text{ V}$$

$$v_x = 20 - 12 = 8 \text{ V}$$

Kirchhoff's Voltage Law (KVL)

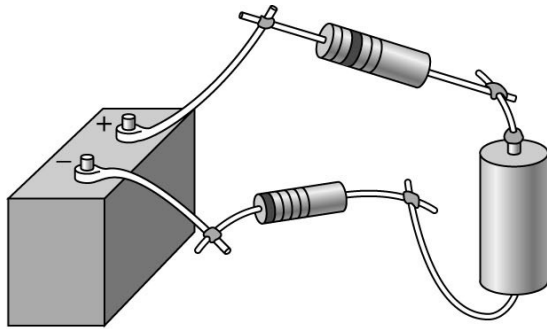
Practice

Determine v_x in the circuit



The Single-Loop Circuit

All of the elements in a circuit that carry the same i are connected **in series**



(a)

We seek:

the current **through** each element

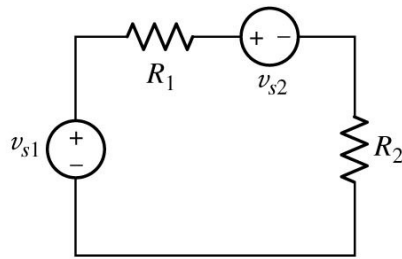
the voltage **across** each element

the power **absorbed** by each element

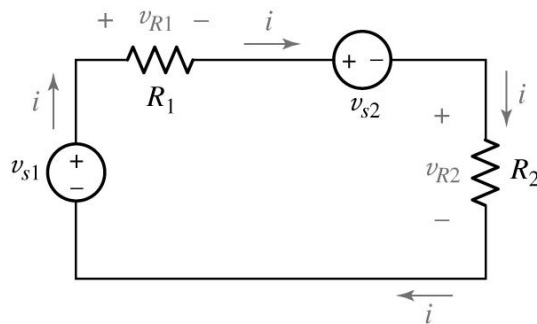
1st step: assign current direction – **the same current everywhere**

2nd step: assign voltages v_{R1} and v_{R2} according with i direction

3rd step: apply KVL



(b)



(c)

$$-v_{s1} + v_{R1} + v_{s2} + v_{R2} = 0$$

$$v_{R1} = R_1 i \quad v_{R2} = R_2 i$$

$$-v_{s1} + R_1 i + v_{s2} + R_2 i = 0$$

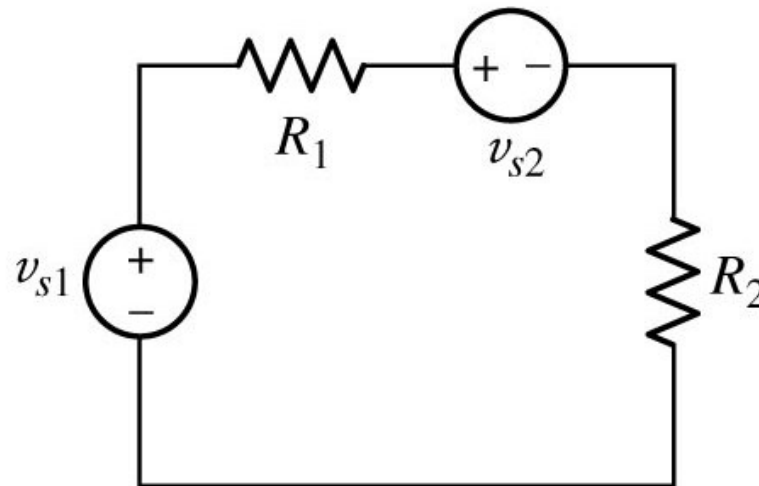
$$i = \frac{v_{s1} - v_{s2}}{R_1 + R_2}$$

Now we can find v and p for every element

The Single-Loop Circuit

Practice

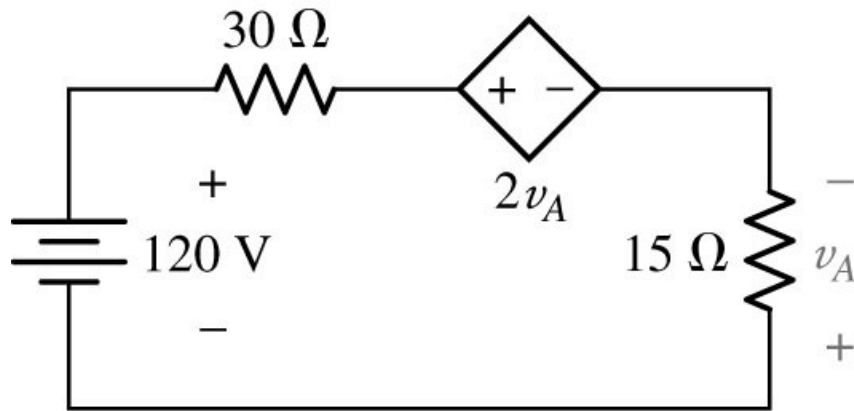
In the circuit $v_{s1} = 120 \text{ V}$, $v_{s2} = 30 \text{ V}$, $R_1 = 30 \Omega$, and $R_2 = 15 \Omega$.
Compute the power absorbed by each element



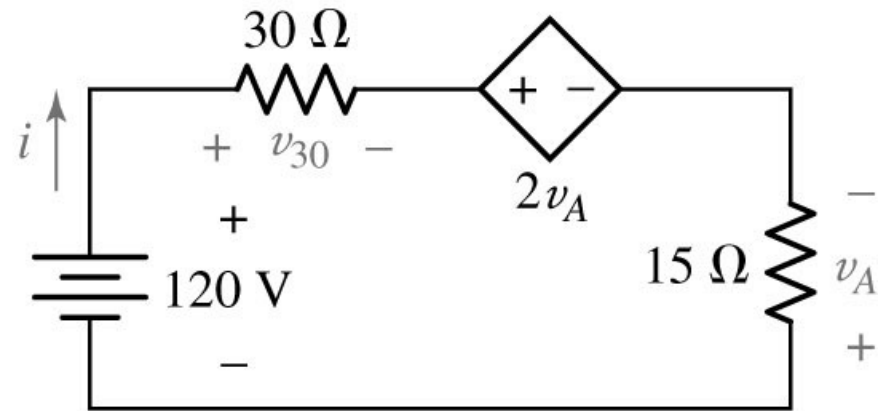
The Single-Loop Circuit

Example

Compute the power absorbed in each element of the circuit



(a)



(b)

KVL: $-120 + v_{30} + 2v_A - v_A = 0$

$$v_{30} = 30i \qquad v_A = -15i$$

$$-120 + 30i - 30i + 15i = 0$$

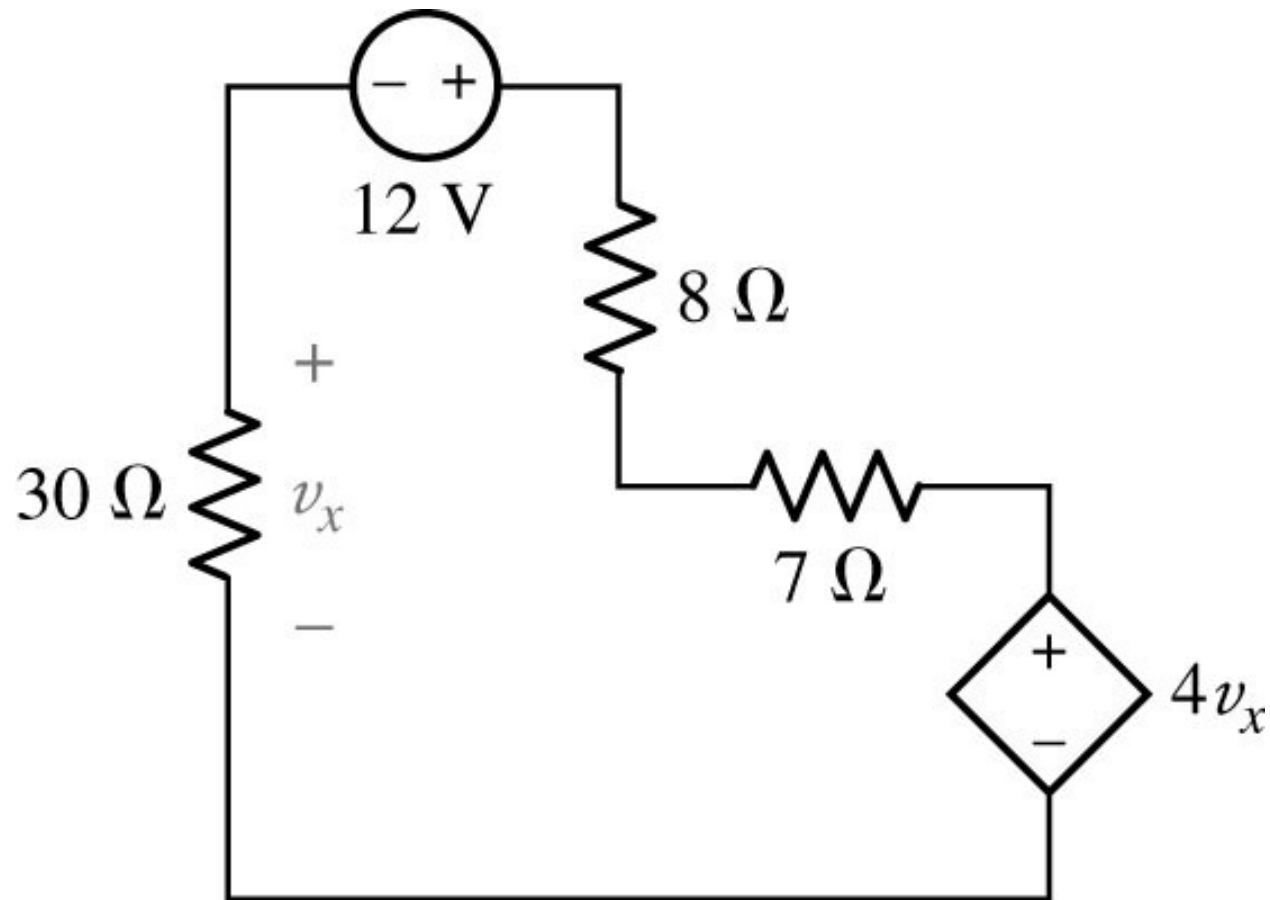
$$i = 8 \text{ A}$$

$$\begin{aligned} P_{120 \text{ V}} &= 120 (-8) = -960 \text{ W} \\ P_{30 \, \Omega} &= (8)^2(30) = 1920 \text{ W} \\ P_{\text{dep}} &= 2[(-15)(8)](8) = -1920 \text{ W} \\ P_{15 \, \Omega} &= (8)^2(15) = 960 \text{ W} \end{aligned}$$

The Single-Loop Circuit

Practice

In the circuit find the power absorbed by each of the five element



The Single-Loop Circuit

The sum of the absorbed power is equal to the sum of the supplied power

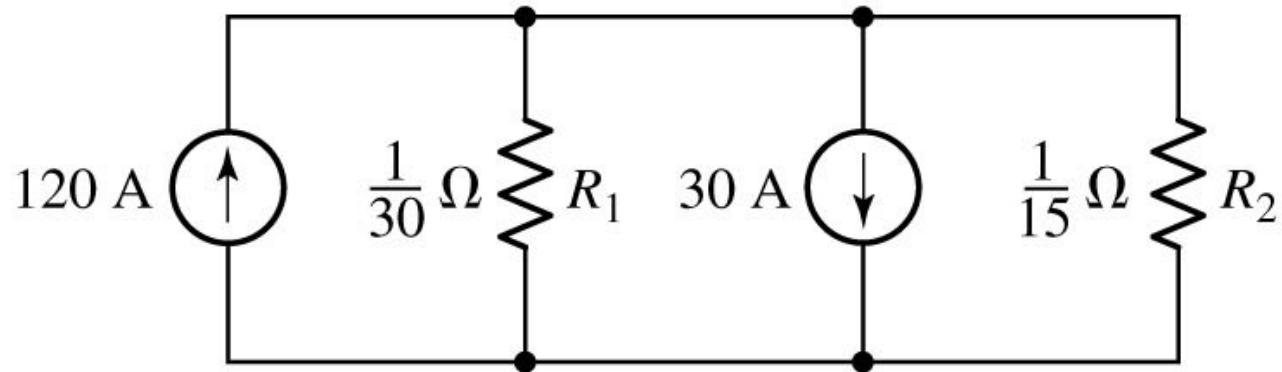
$$\sum P_{Absorbed} = \sum P_{Supplied}$$

$$\sum_{all_elements} P_{absorbed} = -960 + 1920 - 1920 + 960 = 0$$

!!! In Practice problem 12 V source absorbs power

The Single-Node-Pair Circuit

Single-node-pair circuit – a circuit in which any number of single elements are connected between the same pair of nodes

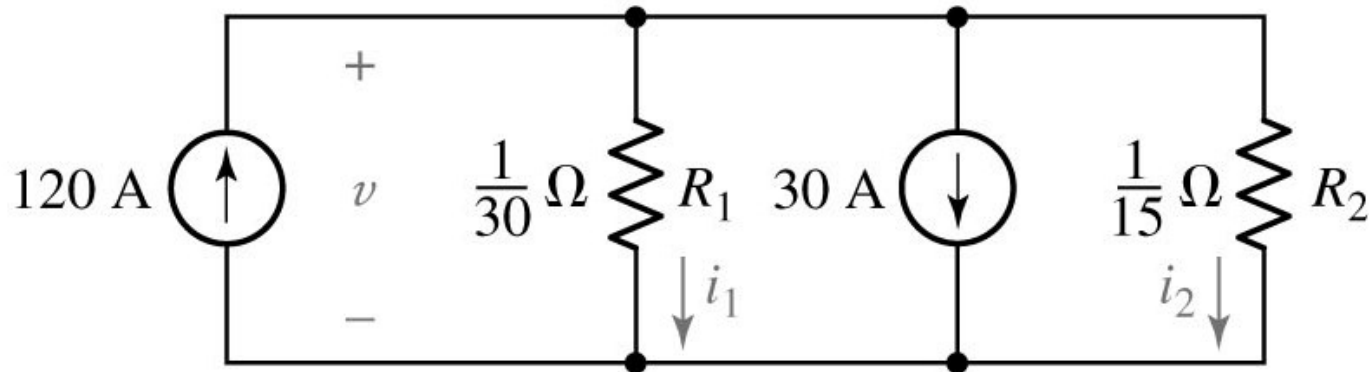
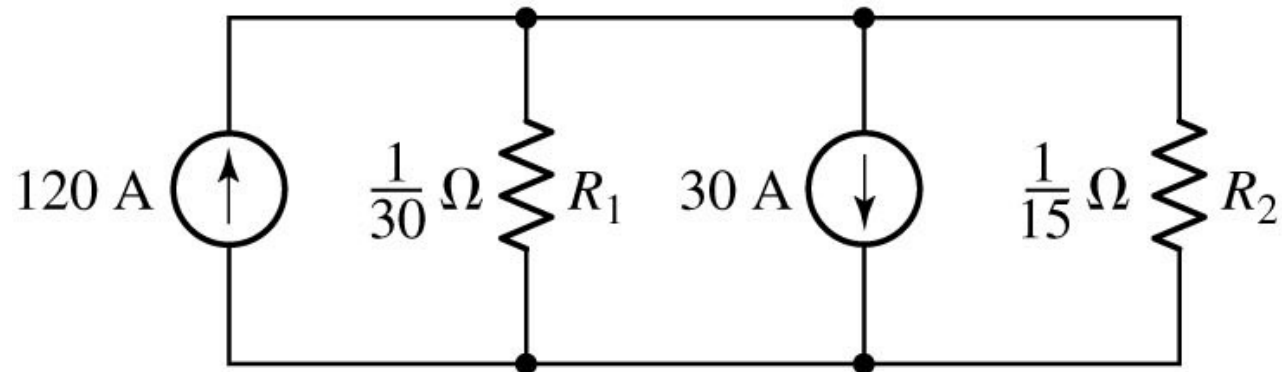


Elements in a circuit having a common voltage across them are said to be ***connected in parallel***

The Single-Node-Pair Circuit

Example

Find the voltage, current, and power associated with each element



$$\text{KCL: } -120 + i_1 + 30 + i_2 = 0$$

$$i_1 = 30v$$

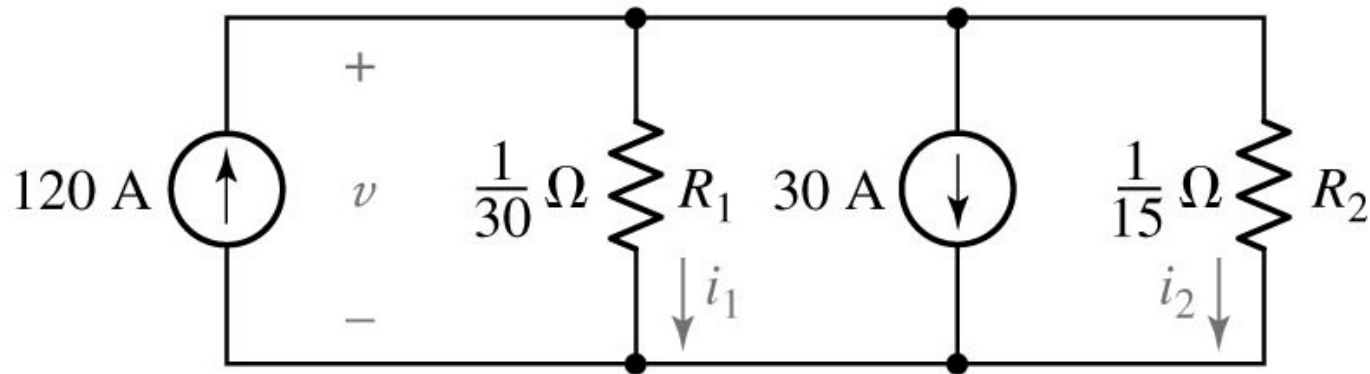
$$i_2 = 15v$$

$$v = 2 \text{ V}$$

The Single-Node-Pair Circuit

Example

Find the voltage, current, and power associated with each element



$$i_1 = 60 \text{ A}$$

$$i_2 = 30 \text{ A}$$

$$p_{R_1} = 30(2)^2 = 120 \text{ W}$$

$$p_{R_2} = 15(2)^2 = 60 \text{ W}$$

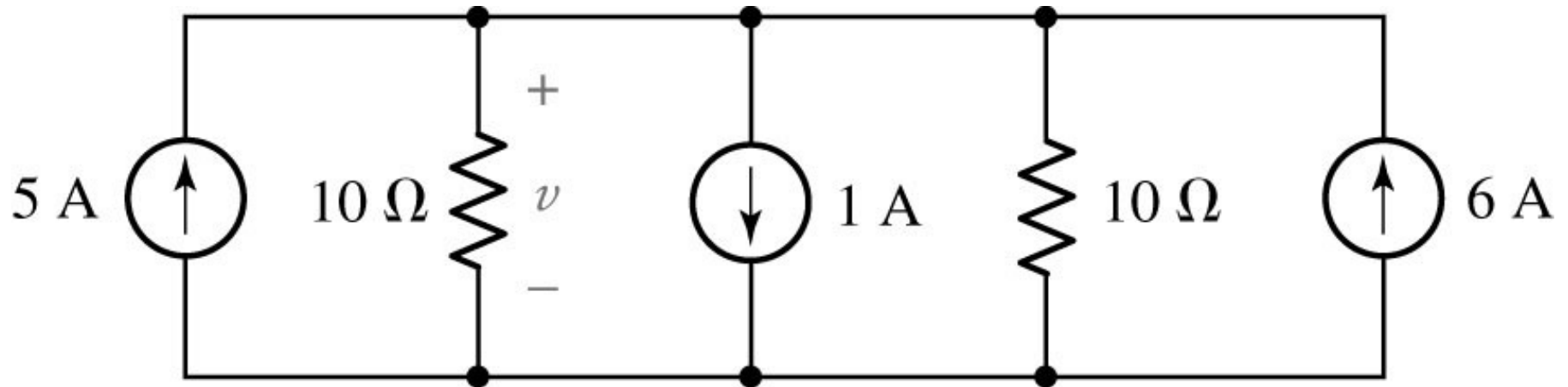
$$P_{120 \text{ A}} = 2(-120) = -240 \text{ W}$$

$$P_{30 \text{ A}} = 30(2) = 60 \text{ W}$$

The Single-Node-Pair Circuit

Practice

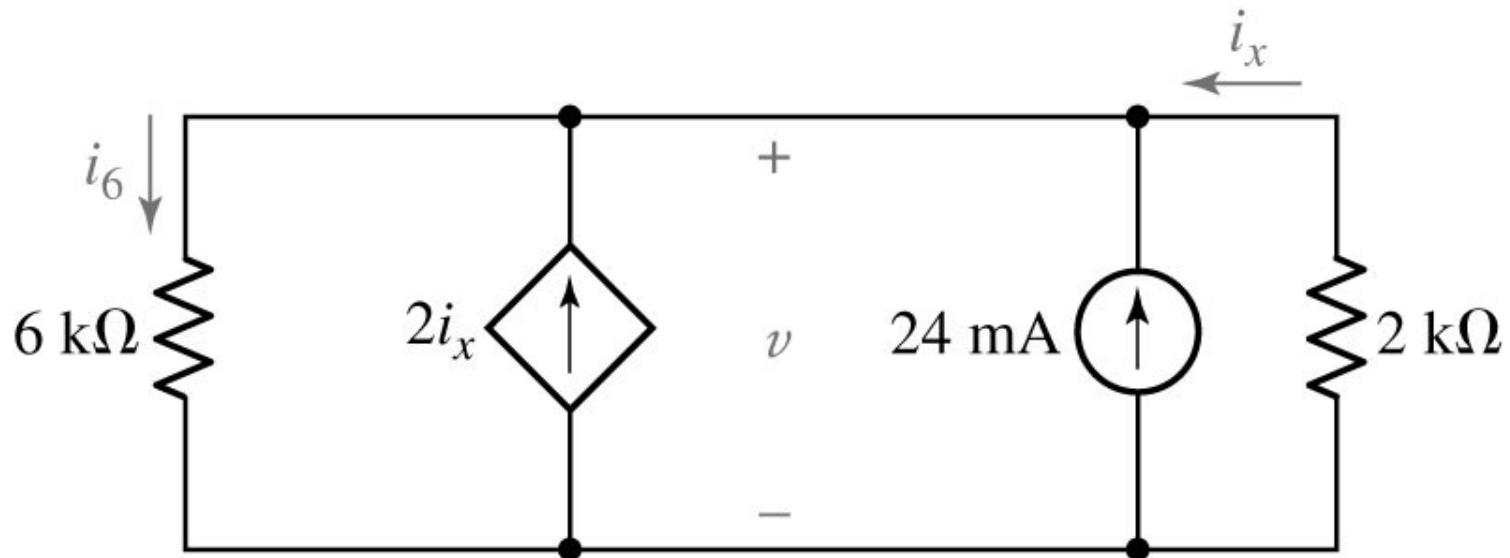
Determine v in the circuit



The Single-Node-Pair Circuit

Example

Determine the value of v and the power supplied by the independent current source



$$\text{KCL: } i_6 - 2i_x - 0.024 - i_x = 0 \quad i_6 = v/6000 \quad i_x = -v/2000$$

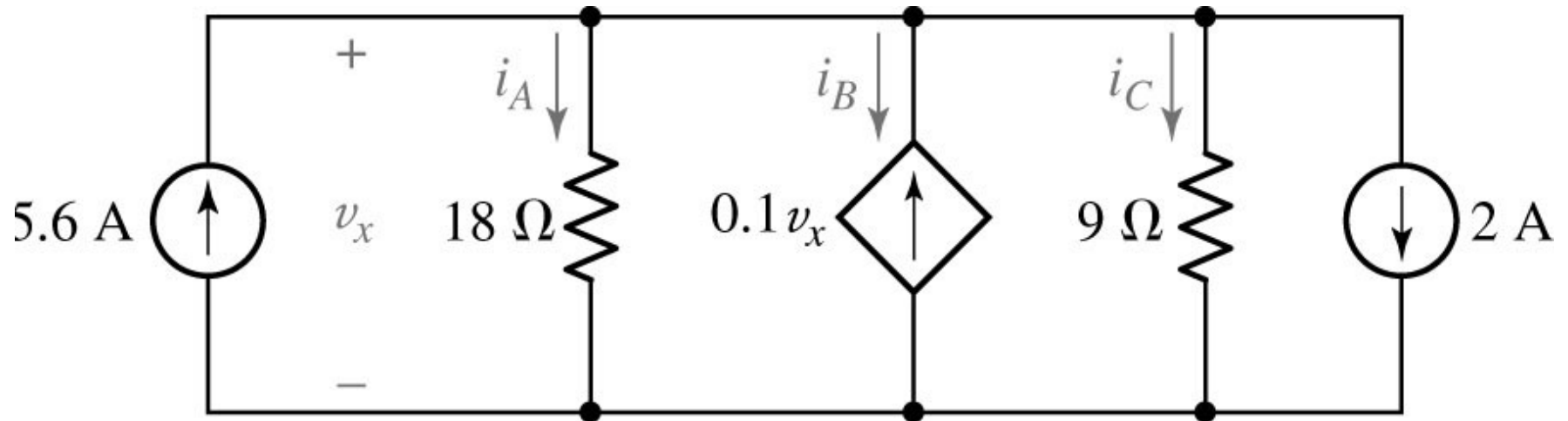
$$v = 14.4 \text{ V}$$

$$p_{24} = 14.4(-0.024) = -0.3456 \text{ W}$$

The Single-Node-Pair Circuit

Practice

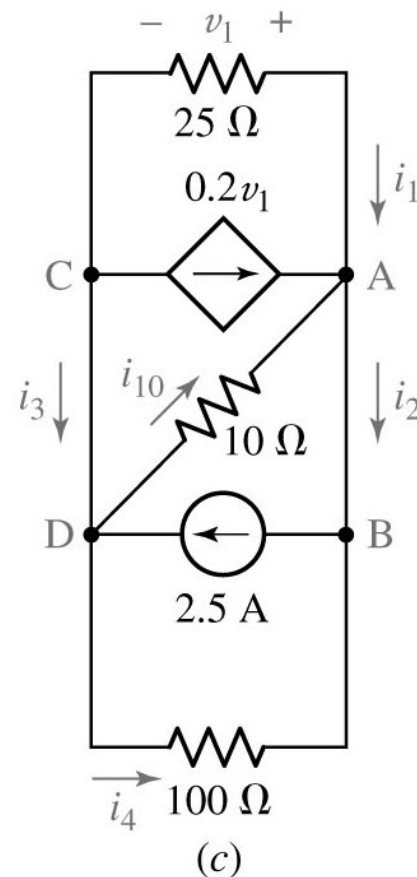
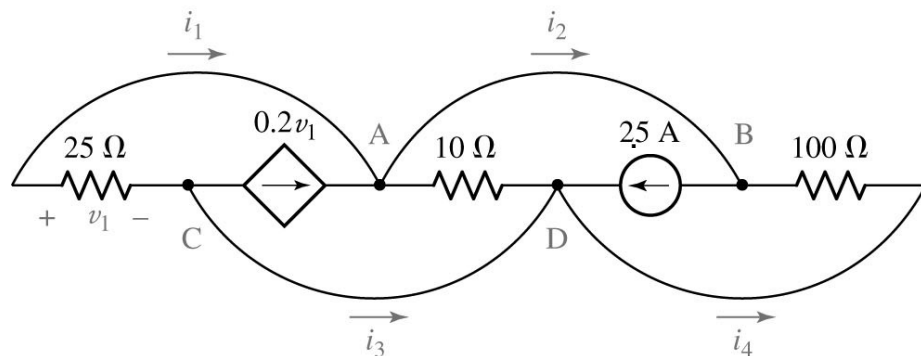
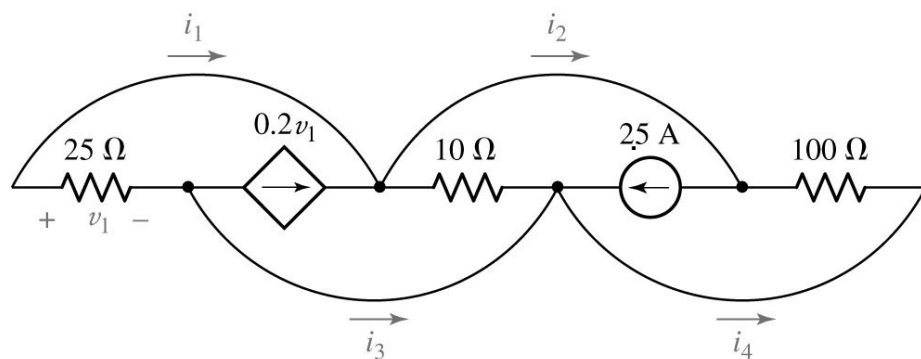
For the single-node-pair circuit find i_A , i_B , and i_C



The Single-Node-Pair Circuit

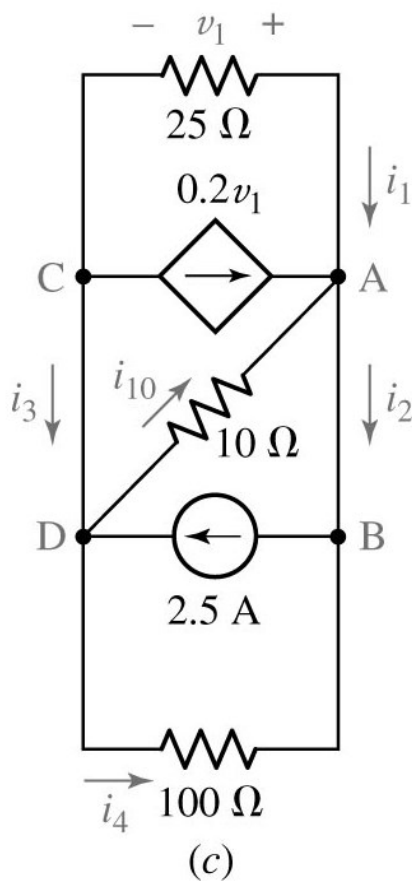
Example

Find i_1 , i_2 , i_3 , and i_4 for the circuit



The Single-Node-Pair Circuit

Example



$$-\frac{v_1}{100} - 2.5 - \frac{v_1}{10} + 0.2v_1 - \frac{v_1}{25} = 0$$

$$v_1 = 50 \text{ V}$$

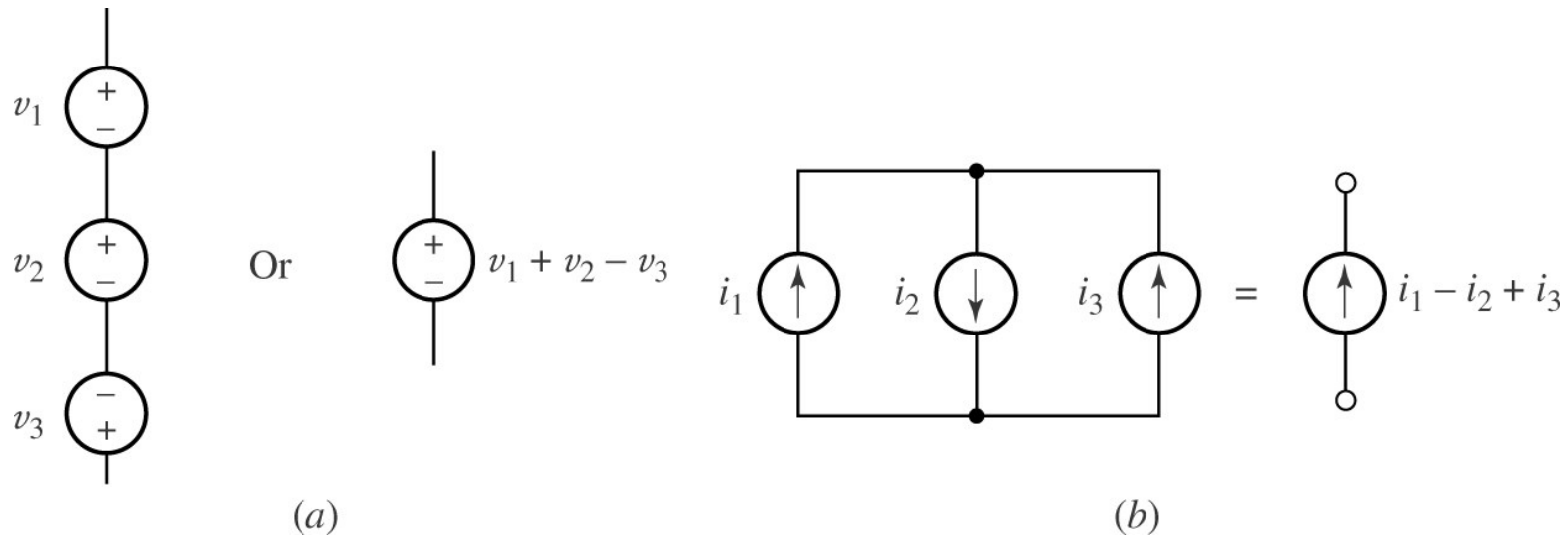
$$i_4 = -\frac{v_1}{100} = -\frac{50}{100} = -0.5 \text{ A}$$

$$i_1 = -2 \text{ A}; i_{10} = -5 \text{ A}$$

$$i_2 = i_1 + 0.2v_1 + i_{10} = -2 + 10 - 5 = 3 \text{ A}$$

$$i_3 = i_{10} - 2.5 + i_4 = -5 - 2.5 - 0.5 = -8 \text{ A}$$

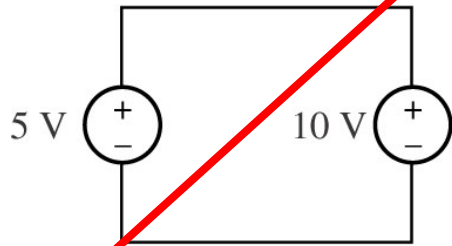
Series and Parallel Connected Sources



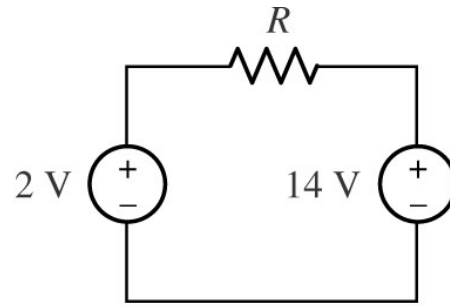
- (a) Series connected voltage sources can be replaced by a single source**
- (b) Parallel current sources can be replaced by a single source.**

Series and Parallel Connected Sources

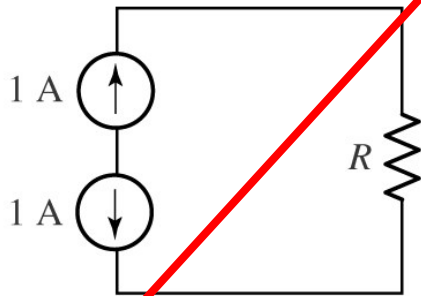
Examples of circuits with multiple sources, some of which are “illegal” as they violate Kirchhoff’s laws.



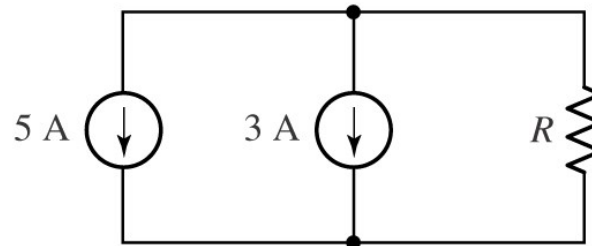
(a)



(b)



(c)

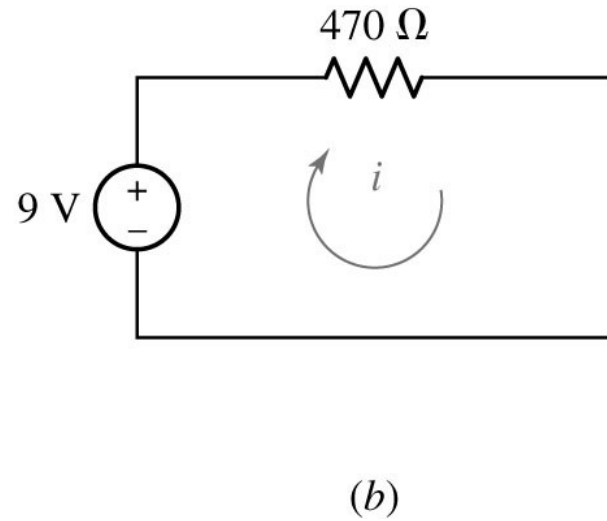
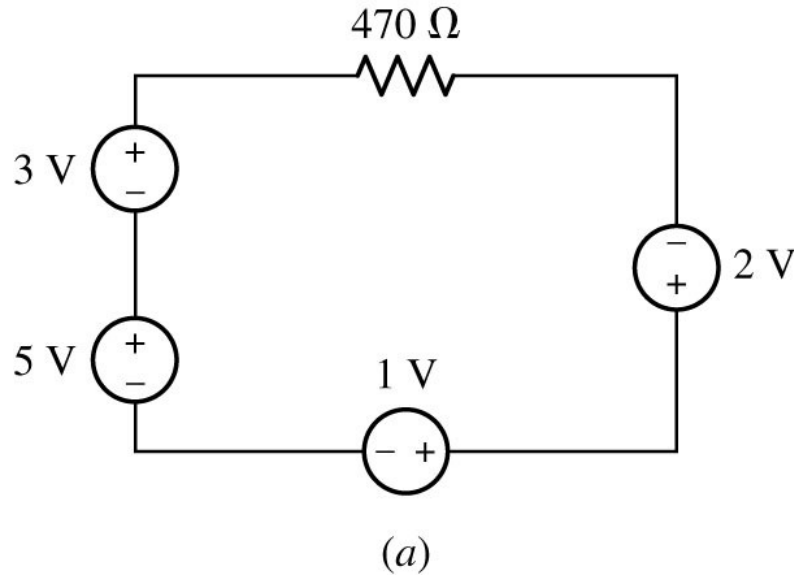


(d)

Series and Parallel Connected Sources

Example

Find the current through the $470\ \Omega$ resistor by first combining the four sources into a single voltage source

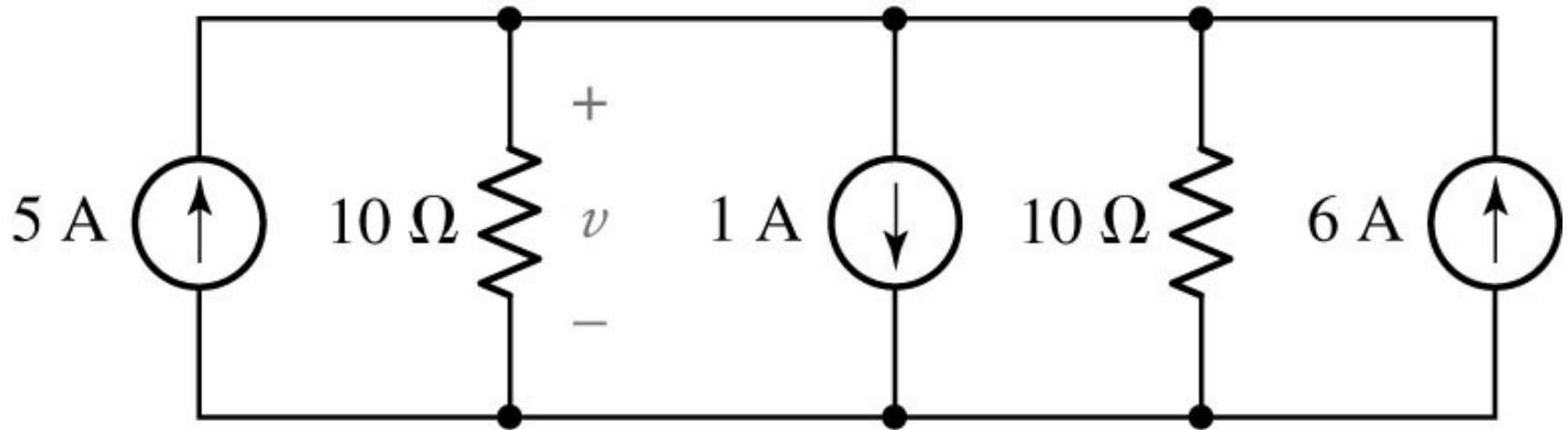


$$i = 9/470 = 19.15\text{ mA}$$

Series and Parallel Connected Sources

Practice

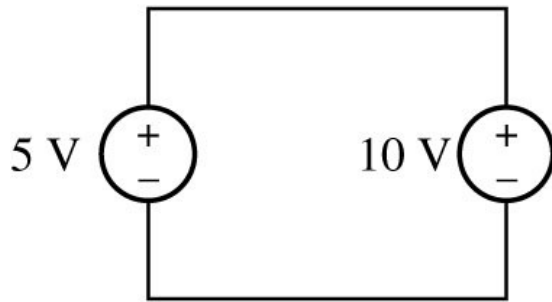
Determine v in the circuit by first combining the three current sources



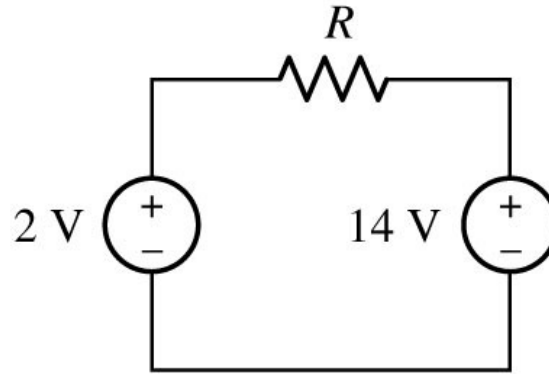
Series and Parallel Connected Sources

Example

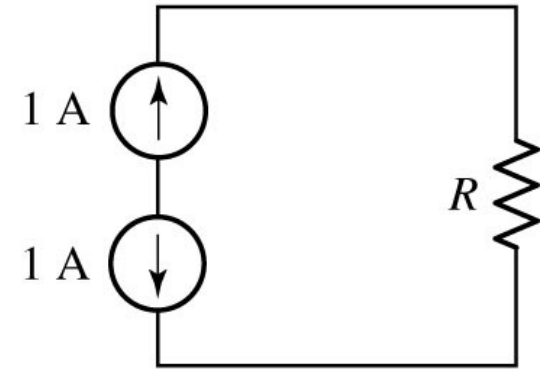
Determine which of the circuits are valid



(a)



(b)

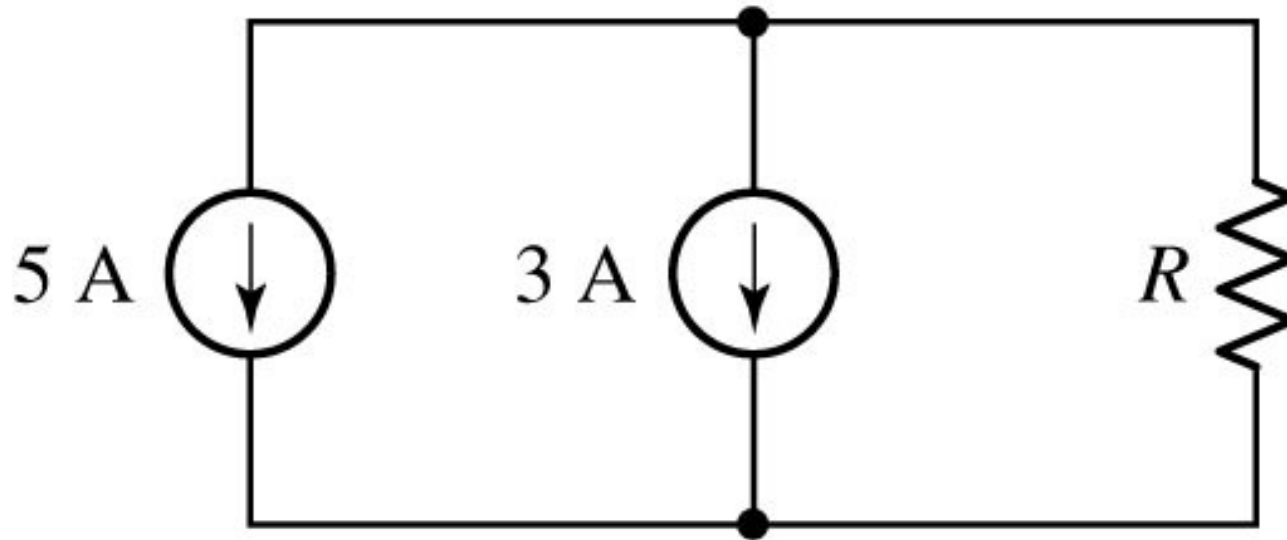


(c)

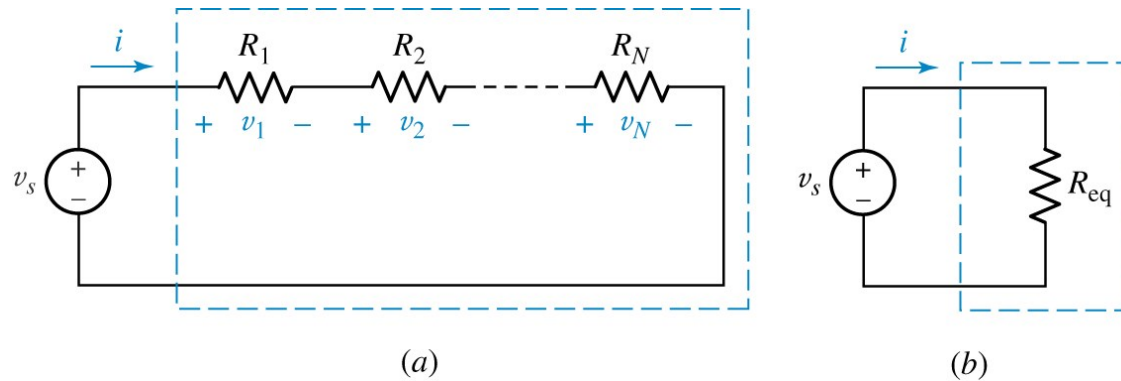
Series and Parallel Connected Sources

Practice

Determine whether the circuit violates either of Kirchhoff's laws



Resistors in Series and Parallel



(a) Series combination of N resistors. (b) Electrically equivalent circuit.

First, apply KVL:

$$v_s = v_1 + v_2 + \cdots + v_N$$

and then Ohm's law:

$$v_s = R_1 i + R_2 i + \cdots + R_N i = (R_1 + R_2 + \cdots + R_N) i$$

Now compare this result with the simple equation applying to the equivalent circuit shown

$$v_s = R_{eq} i$$

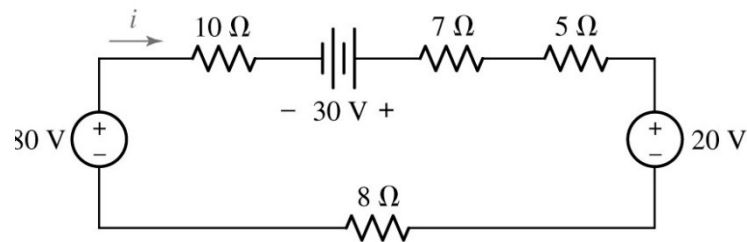
Thus, the value of the equivalent resistance for N series resistors is

$$R_{eq} = R_1 + R_2 + \cdots + R_N$$

Resistors in Series and Parallel

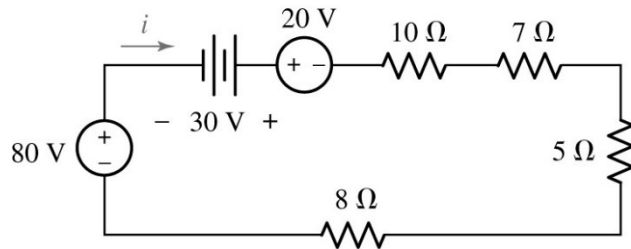
Example

Use resistance and source combinations to determine the current i and the power delivered by the 80 V source



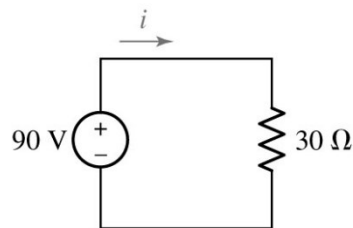
(a)

$$i = 3\text{ A}$$



(b)

$$P = (80)(3) = 240\text{ W}$$

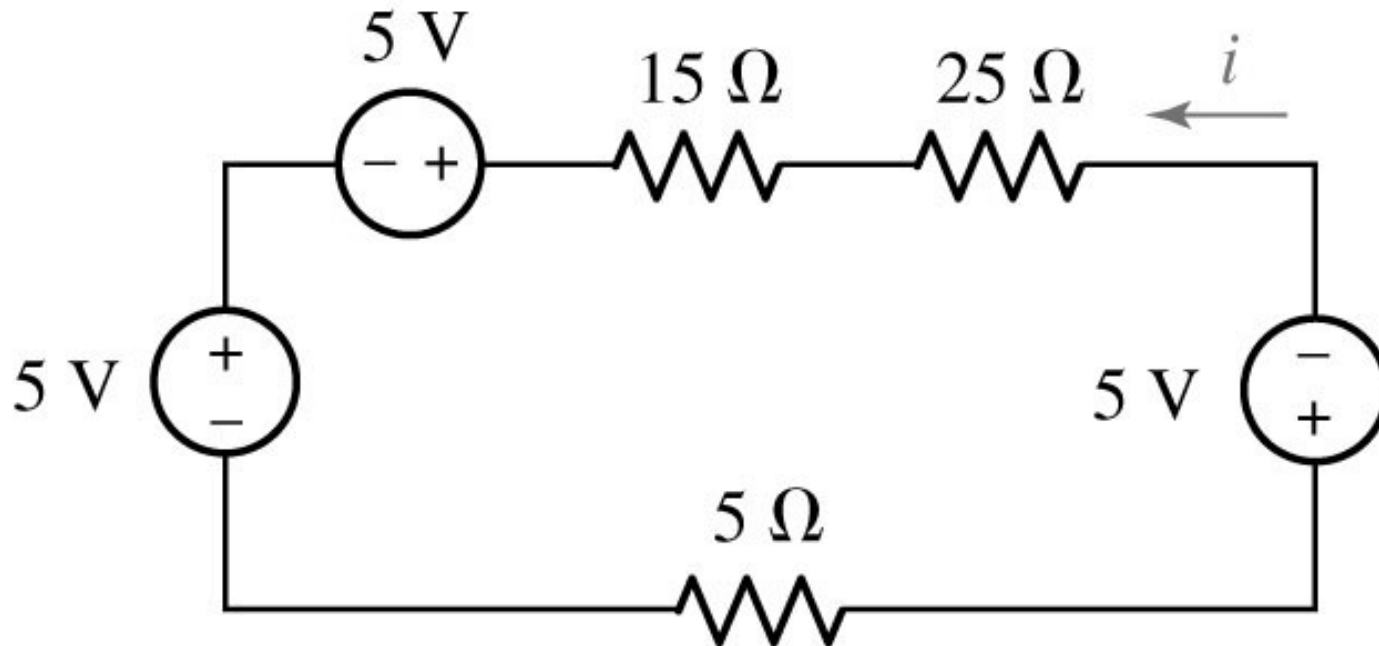


(c)

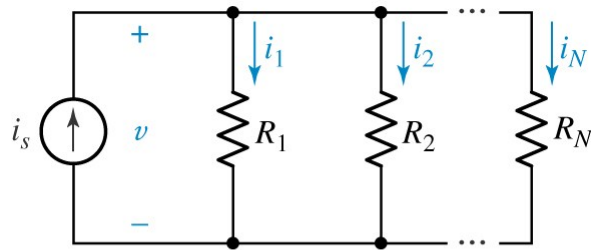
Resistors in Series and Parallel

Practice

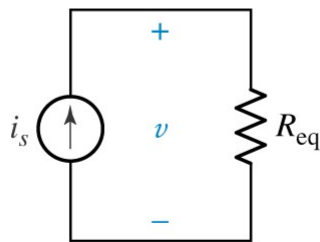
Use resistance and source combinations to determine the current i



Resistors in Series and Parallel



(a)



(b)

(a) A circuit with N resistors in parallel. (b) Equivalent circuit.

Beginning with a simple KCL equation,

$$i_s = i_1 + i_2 + \dots + i_N$$

or
$$i_s = \frac{v}{R_1} + \frac{v}{R_2} + \dots + \frac{v}{R_N} = \frac{v}{R_{eq}}$$

Thus,
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}$$

A special case worth remembering is

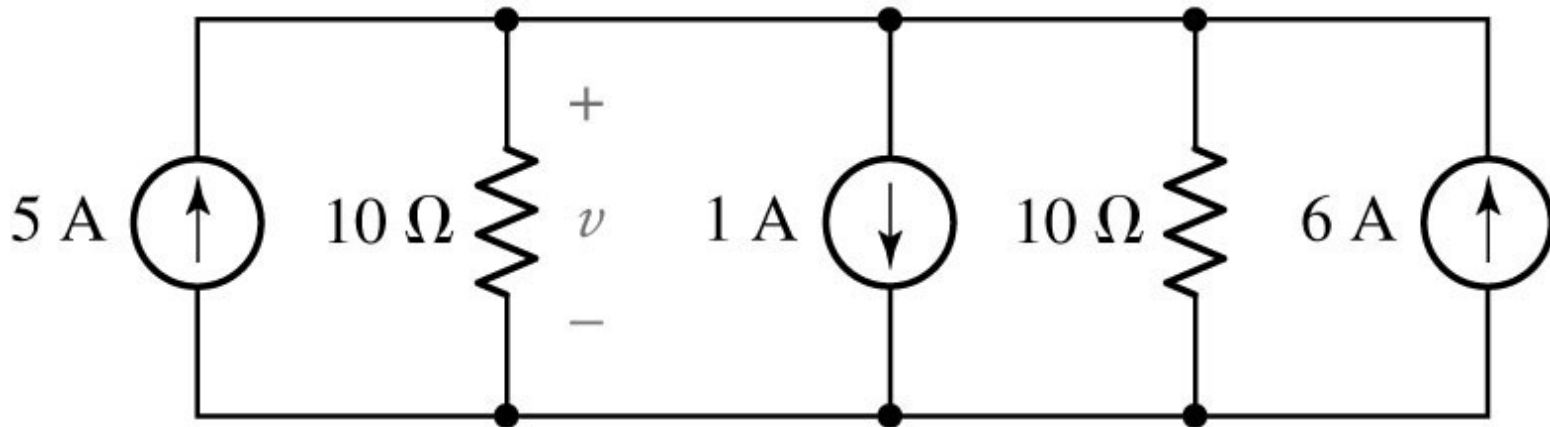
$$R_{eq} = R_1 \parallel R_2 = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Resistors in Series and Parallel

Practice

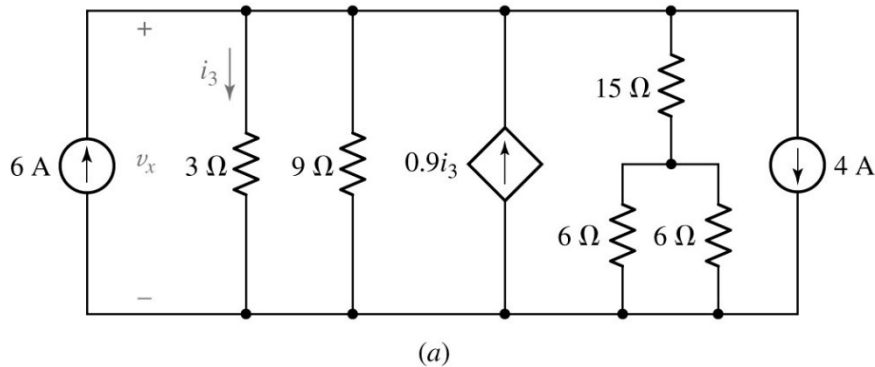
Determine v in the circuit by first combining the three current sources and then the two $10\ \Omega$ resistors



Resistors in Series and Parallel

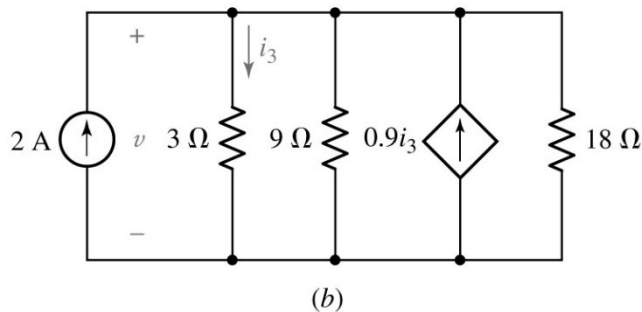
Example

Calculate the power and voltage on the dependent source



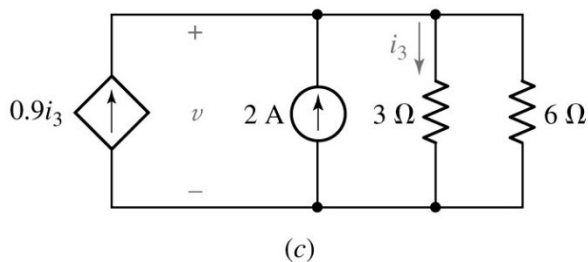
$$\text{KCL: } -0.9i_3 - 2 + i_3 + v/6 = 0$$

$$v = 3i_3$$



$$i_3 = 10/3 \text{ A}$$

$$v = 3i_3 = 10 \text{ V}$$

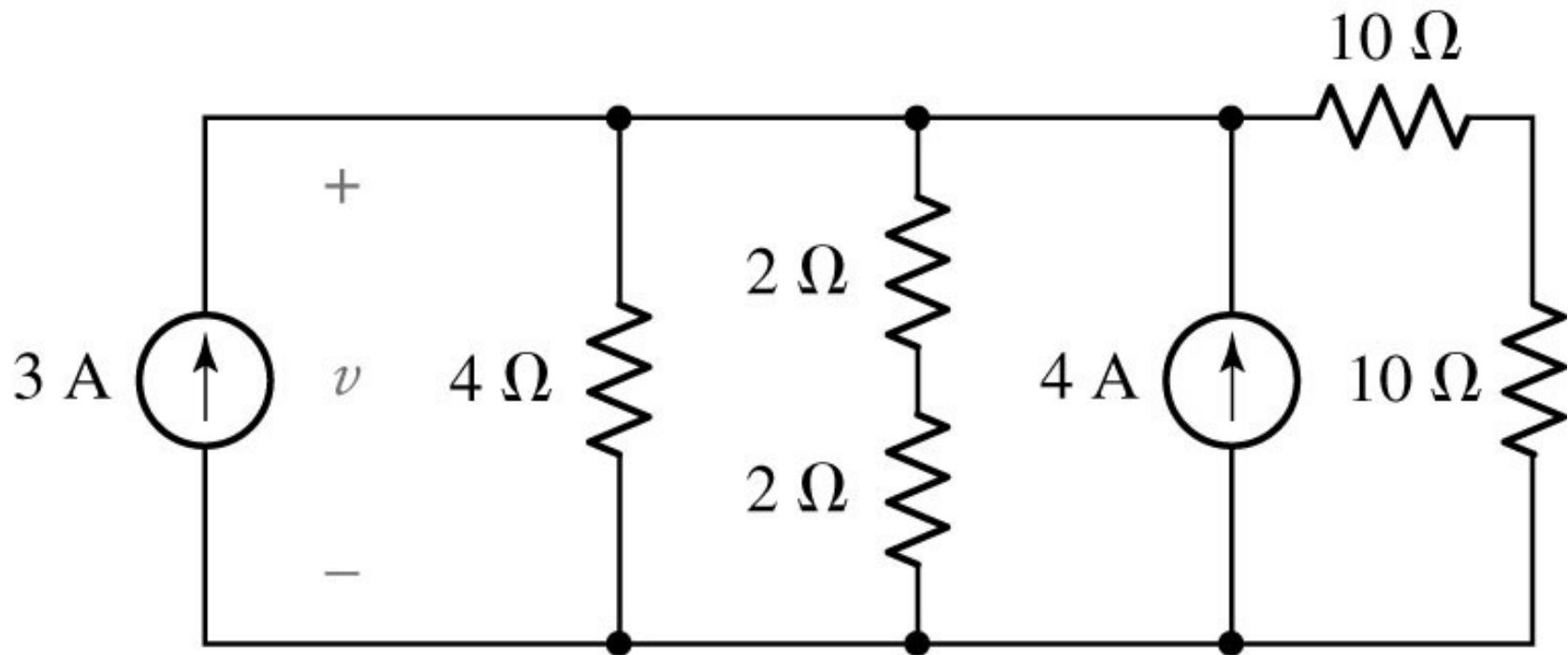


$$P_{0.9i_3} = v \cdot (-0.9i_3) = -30 \text{ W}$$

Resistors in Series and Parallel

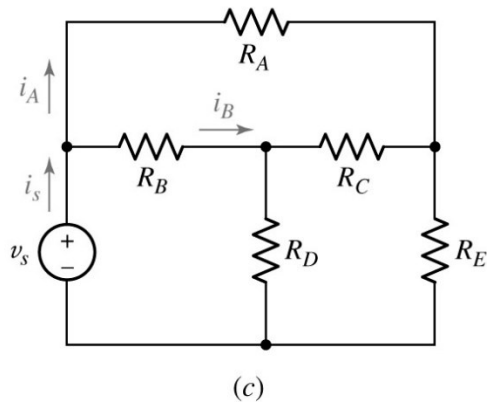
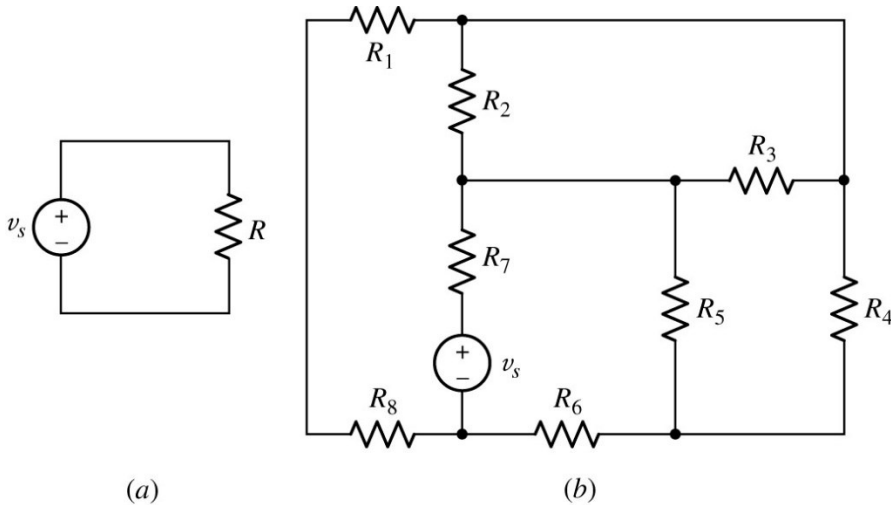
Practice

For the circuit find the voltage v



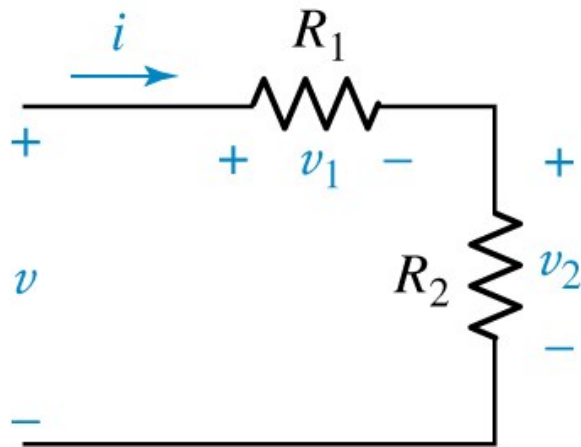
Resistors in Series and Parallel

Example



- a) The two elements are both in series and parallel
- b) R_2 and R_3 are in parallel, and R_1 and R_8 are in series
- c) There are no circuit elements either in series or in parallel with one another

Voltage Division



An illustration of voltage division.

We may find v_2 by applying KVL and Ohm's law:

$$v = v_1 + v_2 = iR_1 + iR_2 = i(R_1 + R_2)$$

so
$$i = \frac{v}{R_1 + R_2}$$

Thus,
$$v_2 = iR_2 = \left(\frac{v}{R_1 + R_2} \right) R_2$$

or
$$v_2 = \frac{R_2}{R_1 + R_2} v$$

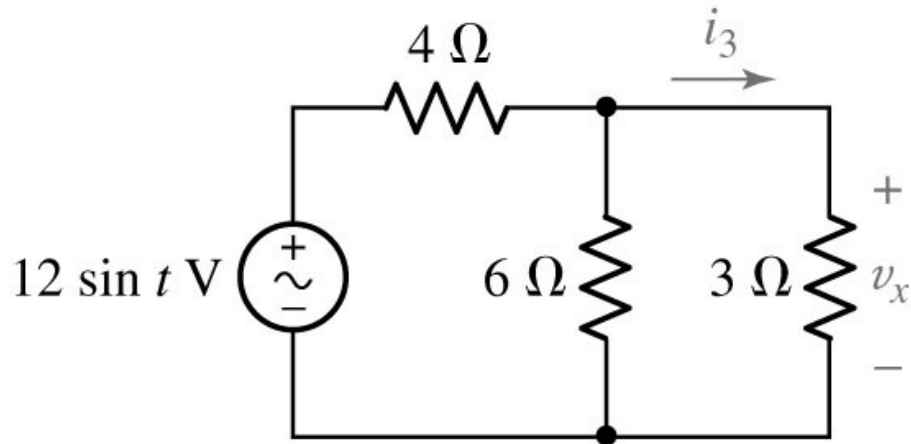
For N resistors in series, we may write:

$$v_k = \frac{R_k}{R_1 + R_2 + \cdots + R_N} v$$

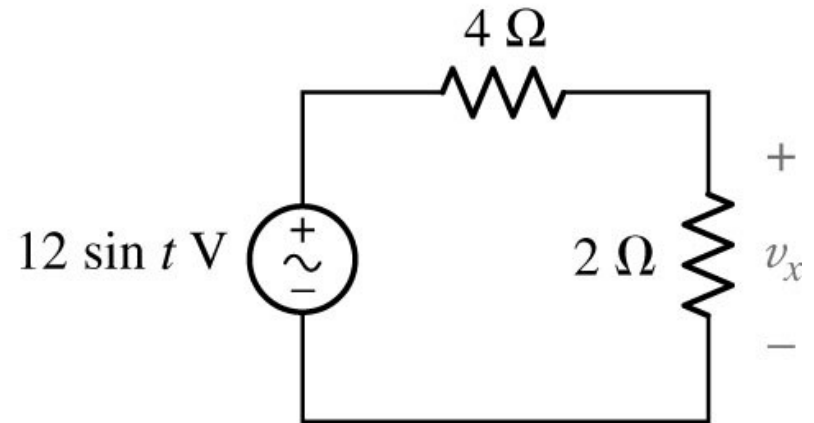
Voltage Division

Example

Determine v_x in the circuit



(a)

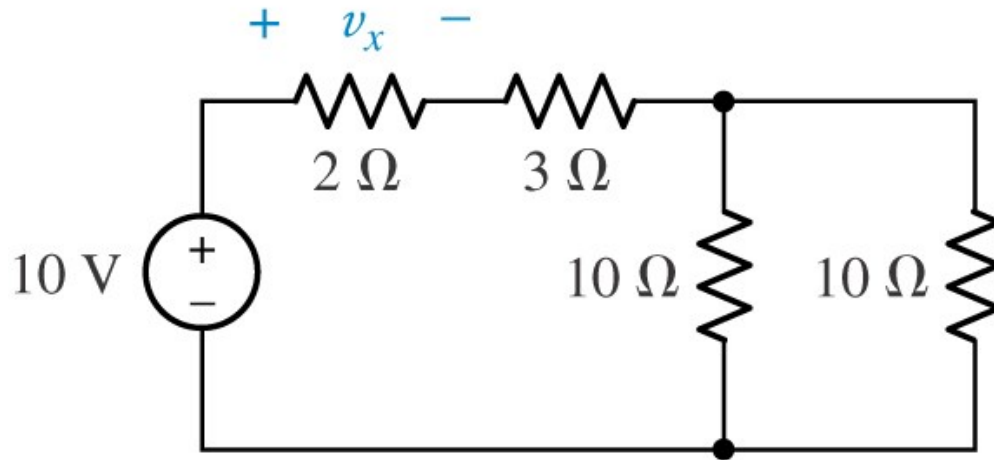


(b)

$$v_x = (12 \sin t) \frac{2}{4 + 2} = 4 \sin t$$

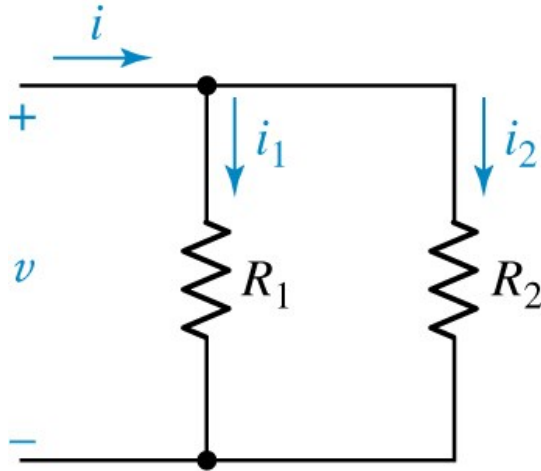
Voltage Division

Practice



Use voltage division to determine v_x in the adjacent circuit.

Current Division



An illustration of current division.

The current flowing through R_2 is

$$i_2 = \frac{v}{R_2} = \frac{i(R_1 \parallel R_2)}{R_2} = \frac{i}{R_2} \frac{R_1 R_2}{R_1 + R_2}$$

or

$$i_2 = i \frac{R_1}{R_1 + R_2}$$

For a parallel combination of N resistors, the current through R_k is

$$i_k = i \frac{\frac{1}{R_k}}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}}$$

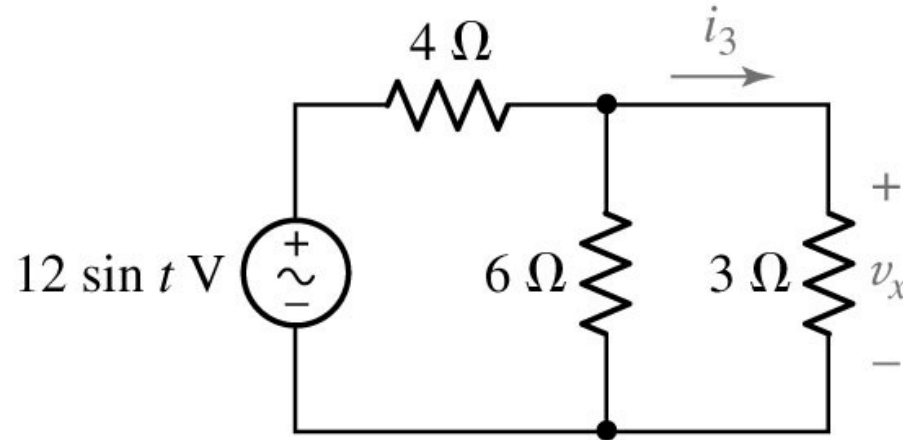
In the terms of conductances:

$$i_k = i \frac{G_k}{G_1 + G_2 + \dots + G_N}$$

Current Division

Example

Write an expression for the current through the $3\ \Omega$ resistance in the circuit



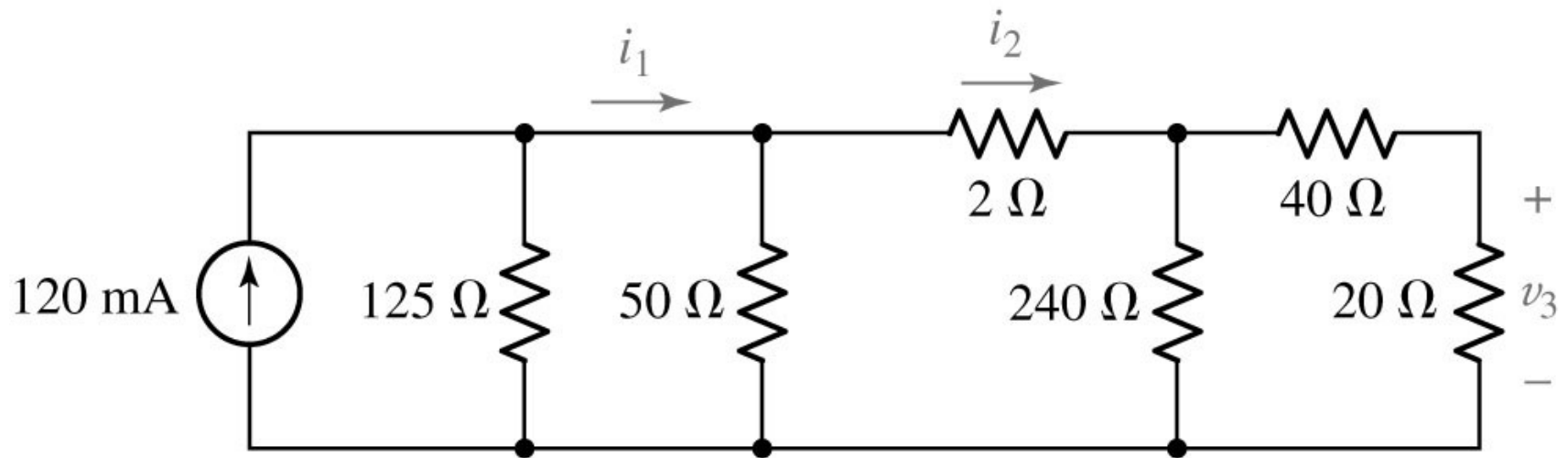
$$i(t) = \frac{12 \sin t}{4 + 3 \parallel 6} = \frac{12 \sin t}{4 + 2} = 2 \sin t$$

$$i_3(t) = (2 \sin t) \left(\frac{1/3}{1/3 + 1/6} \right) = \frac{4}{3} \sin t$$

Current Division

Practice

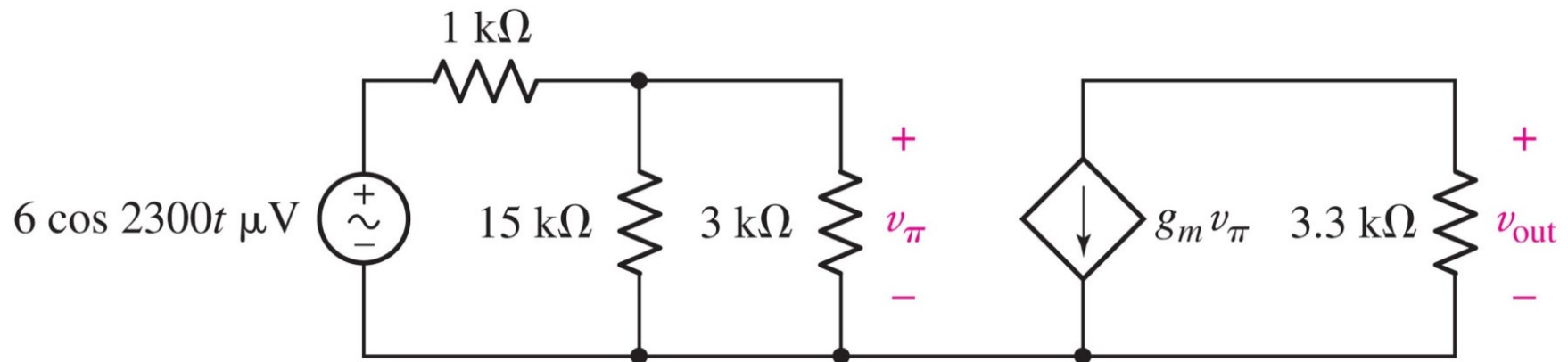
In the circuit, use resistance combination methods and current division to find i_1 , i_2 and v_3 .



Current Division

Practice

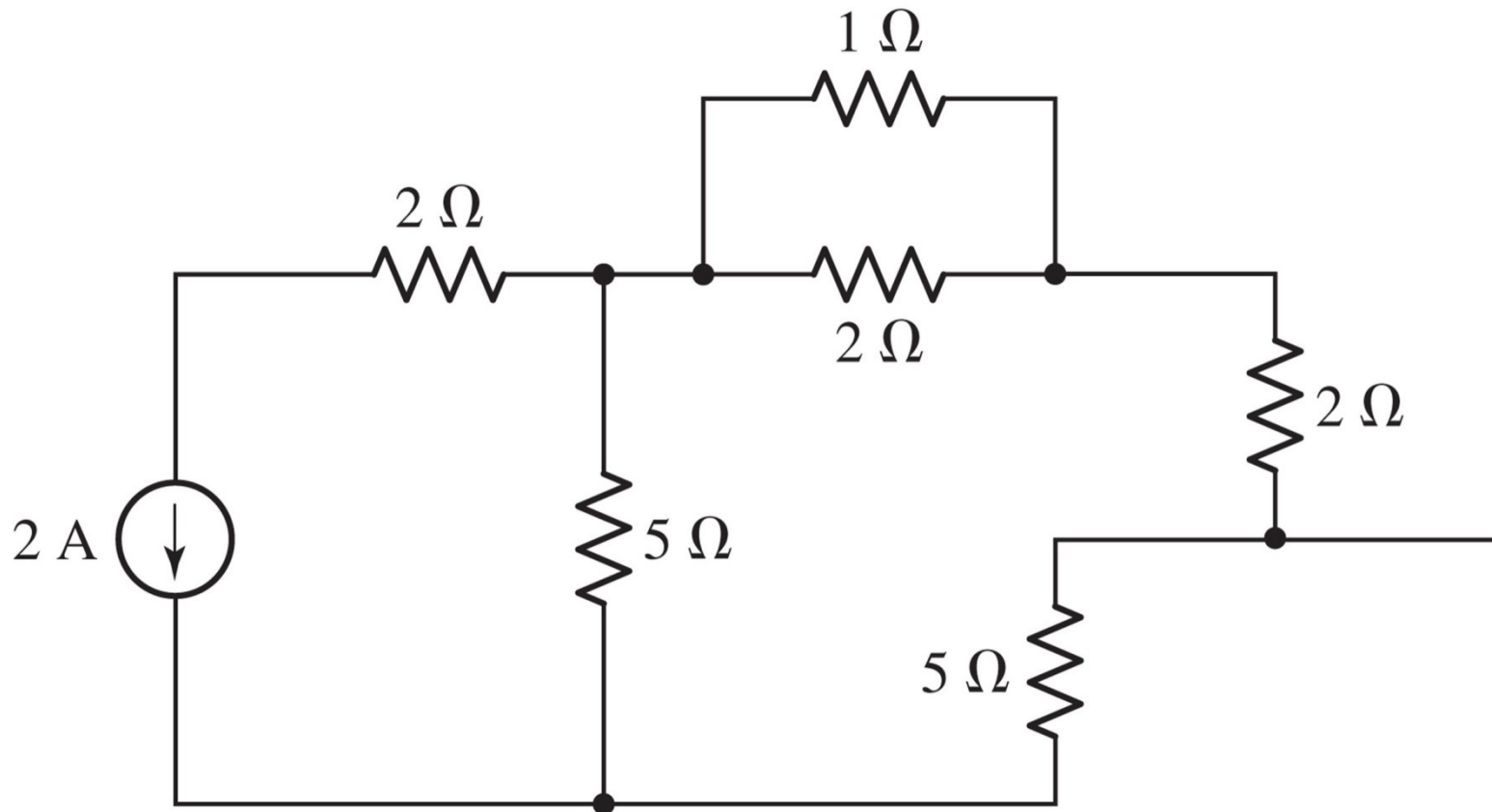
In the circuit, which is a model of bipolar junction transistor-based amplifier, calculate the amplifier output v_{out} , if the transconductance $g_m = 322 \text{ mS}$.



Current Division

Practice

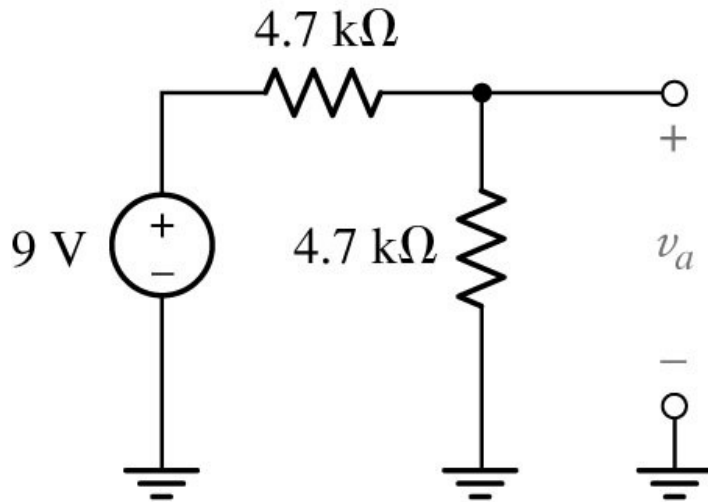
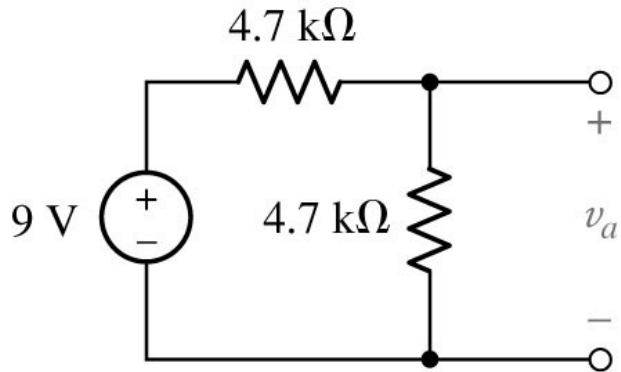
The circuit below: (a) how many nodes, loops, and branches does it contain? (b) Calculate the current flowing through each resistor. (c) Determine the voltage across the current source, assuming the top terminal is the positive reference terminal.



SUMMARY to Chapter 3

1. KCL states that the algebraic sum of the currents entering any node is zero
2. KVL states that the algebraic sum of the voltages around any closed path in a circuit is zero
3. All elements in a circuit that carry the same current are said to be connected in series
4. Elements in a circuit having common voltage across them are said to be connected in parallel
5. A series combination of N resistors can be replaced by a single resistor having the value $R_{eq} = R_1 + R_2 + \dots + R_N$
6. Voltage sources in series can be replaced by a single source, provided care is taken to note the individual polarity of each source
7. Current sources in parallel can be replaced by a single source, provided care is taken to note the direction of each current arrow
8. Voltage division allows to calculate what fraction of the total voltage a series string of resistors is dropped across any one resistor (or group)
9. Current division allows to calculate what fraction of the total current into a parallel string of resistors flows through any one of the resistors

Grounding



(a)

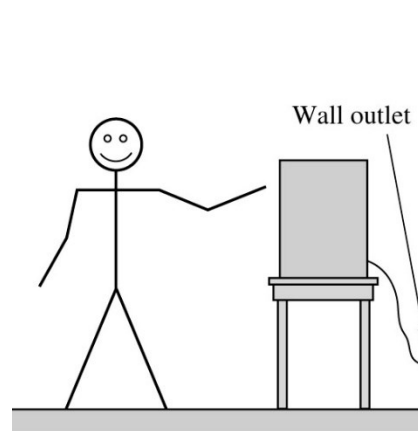


(b)

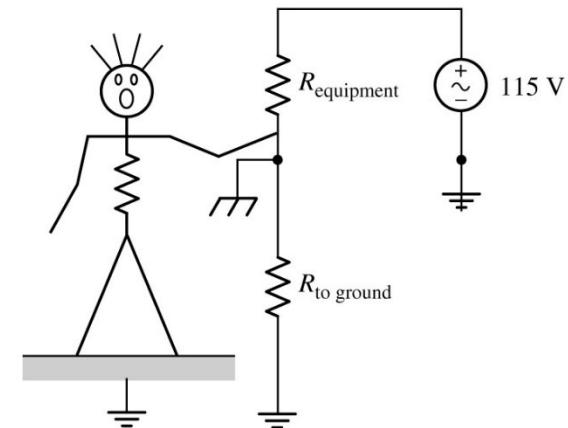


(c)

- a) earth ground
- b) signal ground
- c) chassis ground



(a)



(b)

Diodes

<https://www.youtube.com/watch?v=gkjCe0r5-cw>

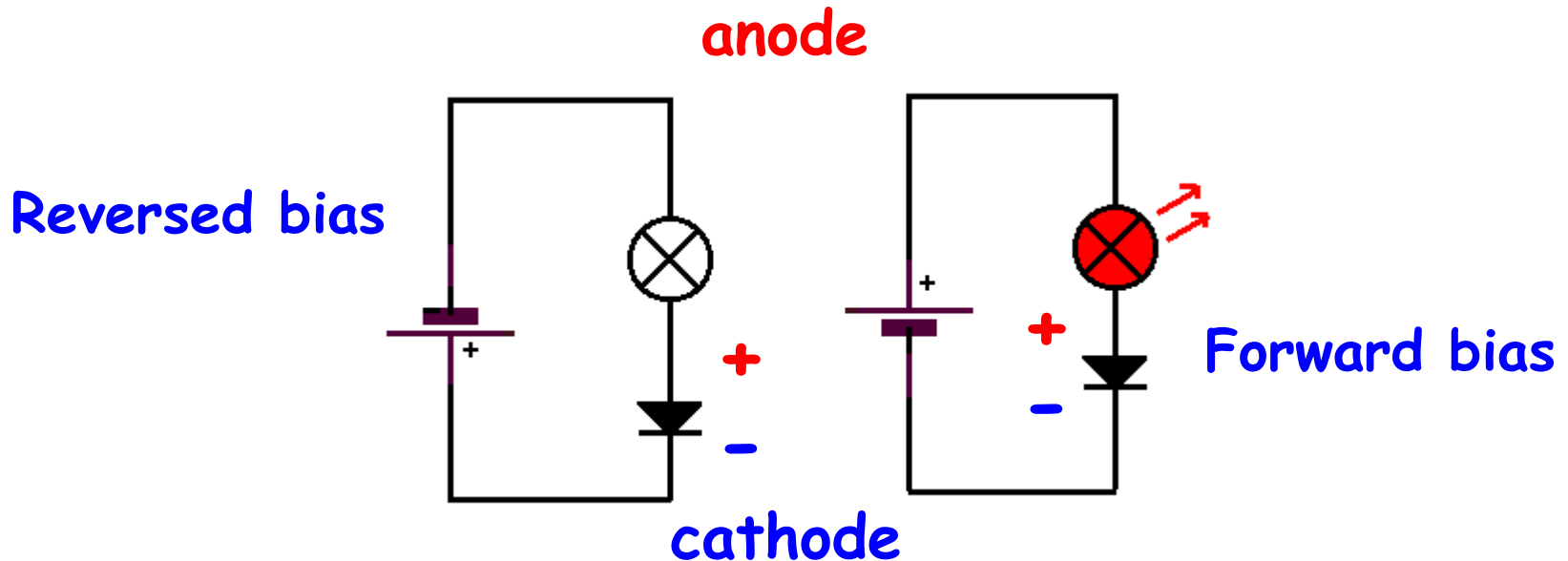
<https://www.youtube.com/watch?v=AgkQrCeJF1Y>

<https://www.youtube.com/watch?v=U8daujO20nM>

<https://www.youtube.com/watch?v=HzwGOZ1QBXU&list=PL9P80IAEKoGT7M1DyMo0K8FwhPbjQz7tt&index=3>

<https://www.youtube.com/watch?v=4SlfaocMfdA&list=PL9P80IAEKoGT7M1DyMo0K8FwhPbjQz7tt>

Practical aspects



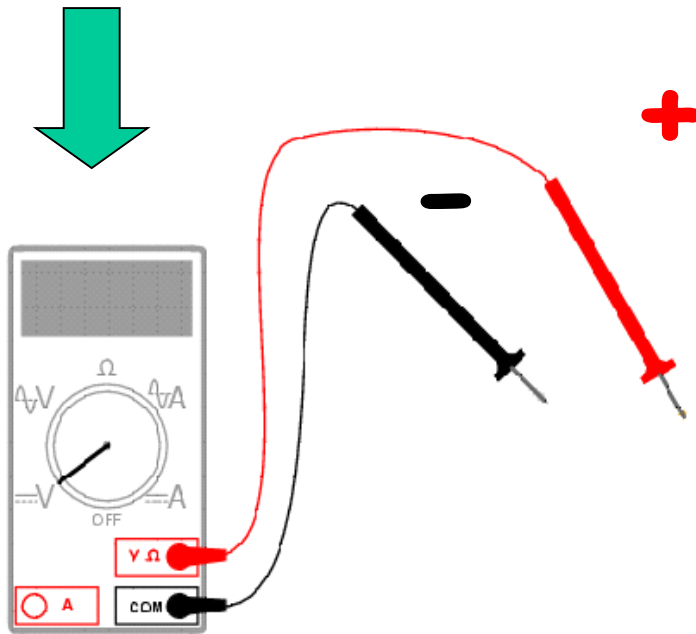
The left hand diagram shows **reverse bias**, with positive on the cathode and negative on the anode (via the lamp). **No current flows.**

The other diagram shows **forward bias**, with positive on the anode and negative on the cathode. **A current flows.**

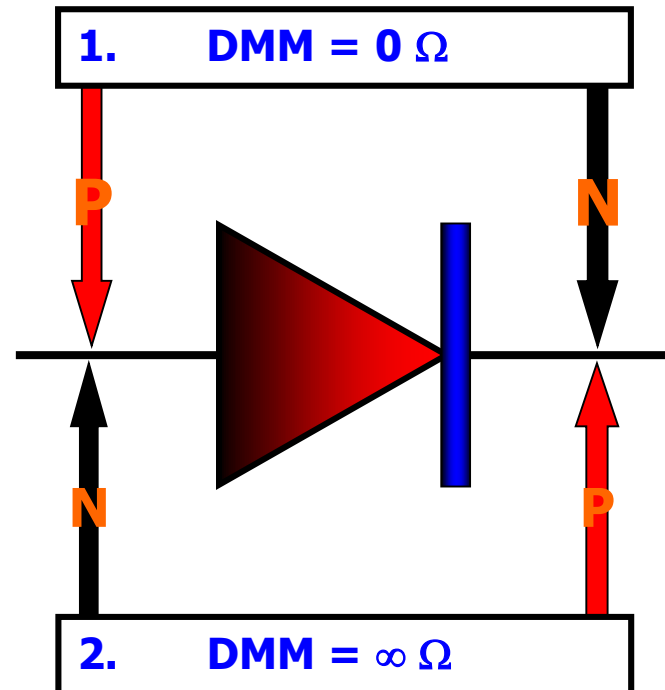
Practical aspects

Diode Ohms Check:

Check by reversing the leads on the Digital Voltage Mutimeter (DMM).



DMM

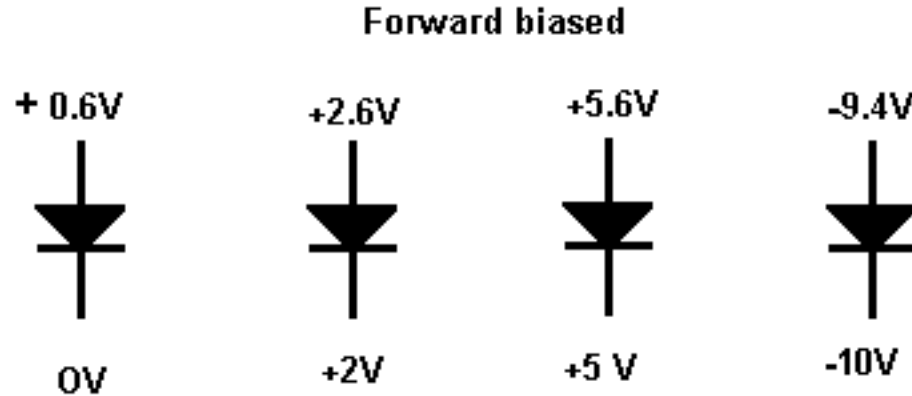


Diodes

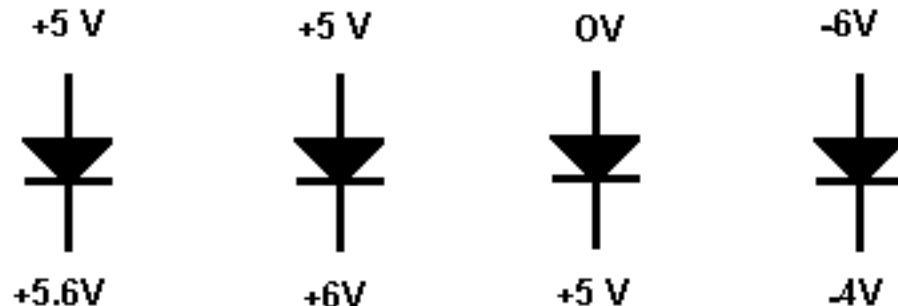
Practical aspects

Diode Voltages

To forward bias a diode, the anode must be more positive than the cathode or **LESS NEGATIVE**.



To reverse bias a diode, the anode must be less positive than the cathode or **MORE NEGATIVE**.

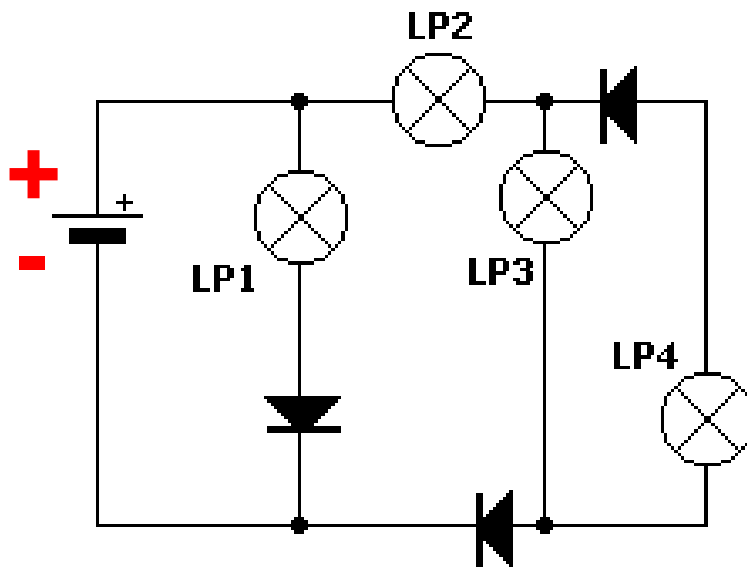


Reverse biased

A conducting diode has about 0.6 volts across if silicon, 0.3 volts if germanium.

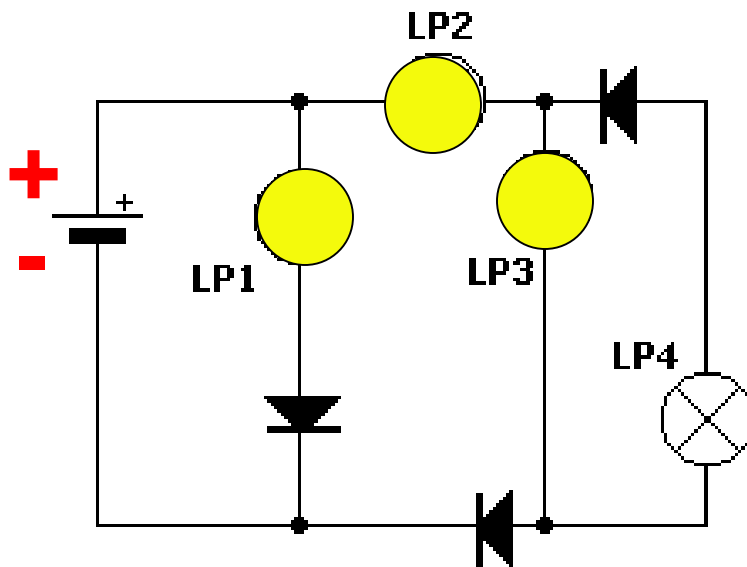
A Diode Puzzle

Which lamps are alight? Some may not be full brightness.



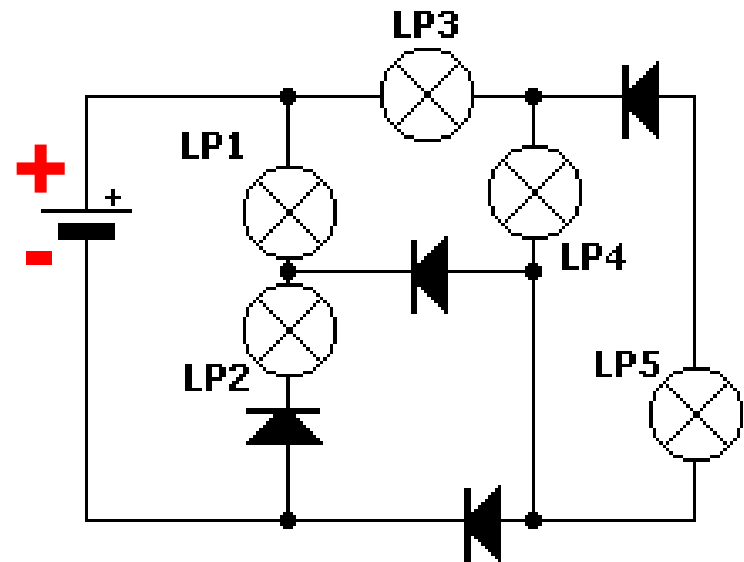
A Diode Puzzle

Which lamps are alight? Some may not be full brightness.



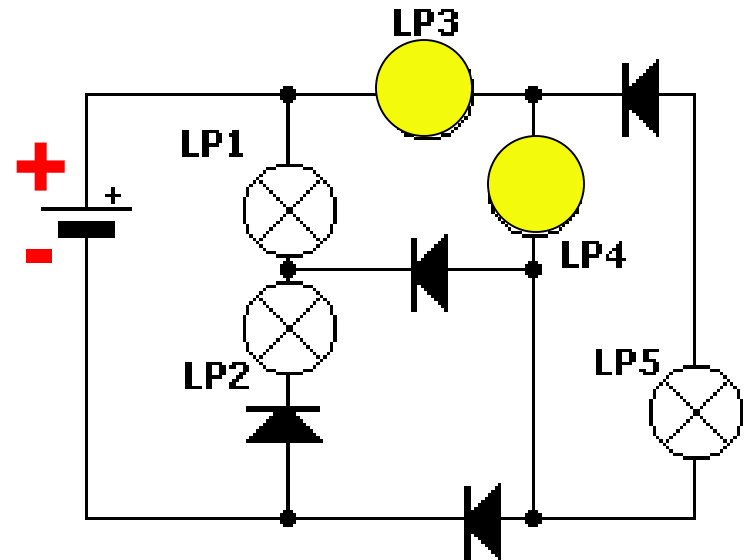
Exercise - a Diode Puzzle

Which lamps are alight? Some may not be full brightness.

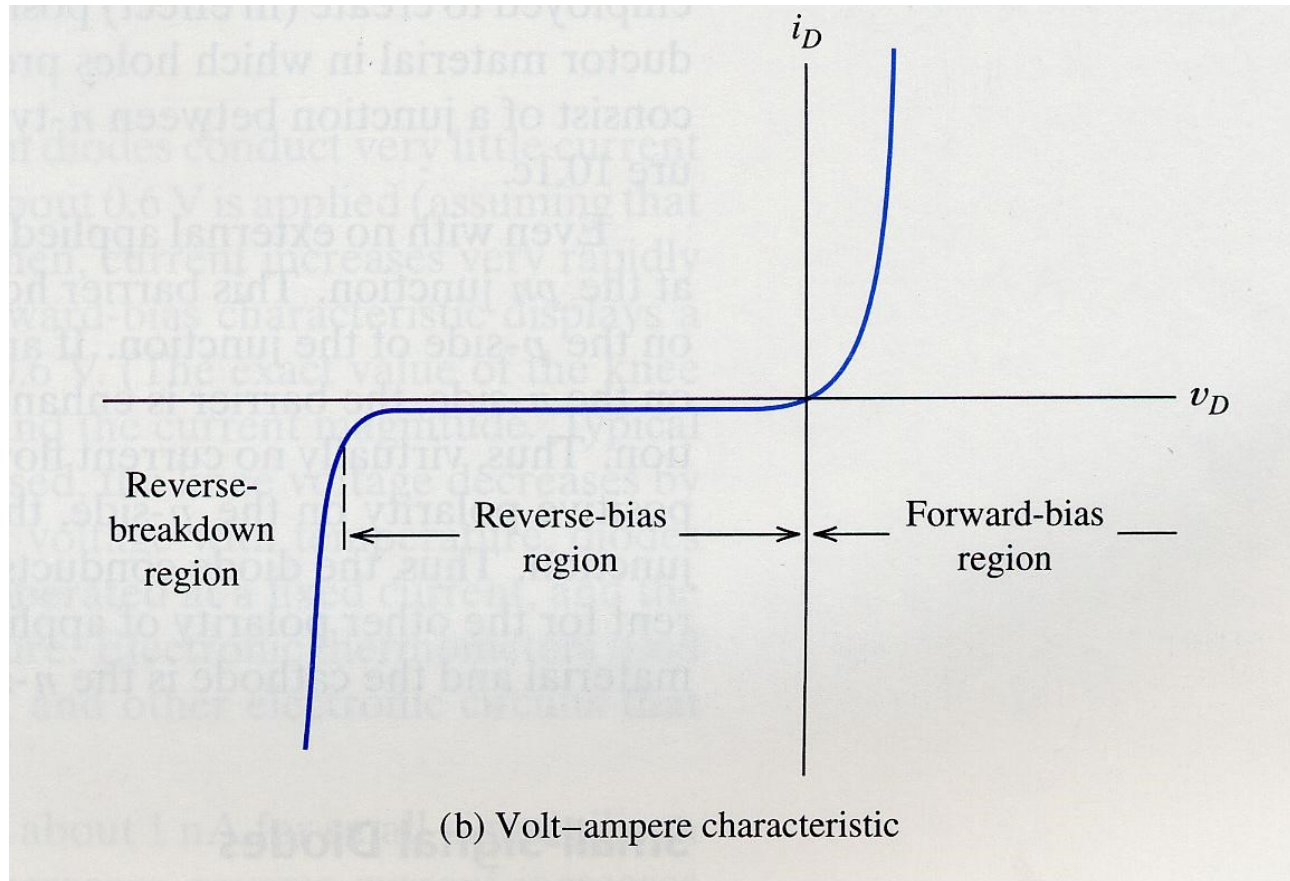


Exercise - a Diode Puzzle

Which lamps are alight? Some may not be full brightness.

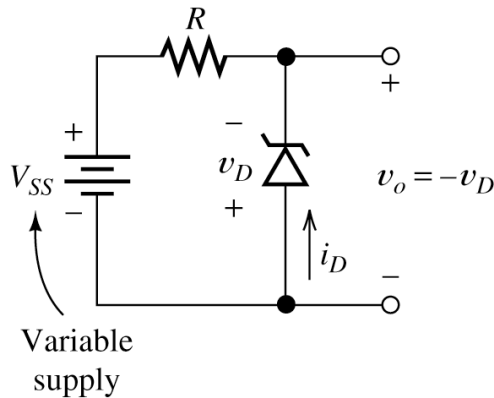


Practical aspects

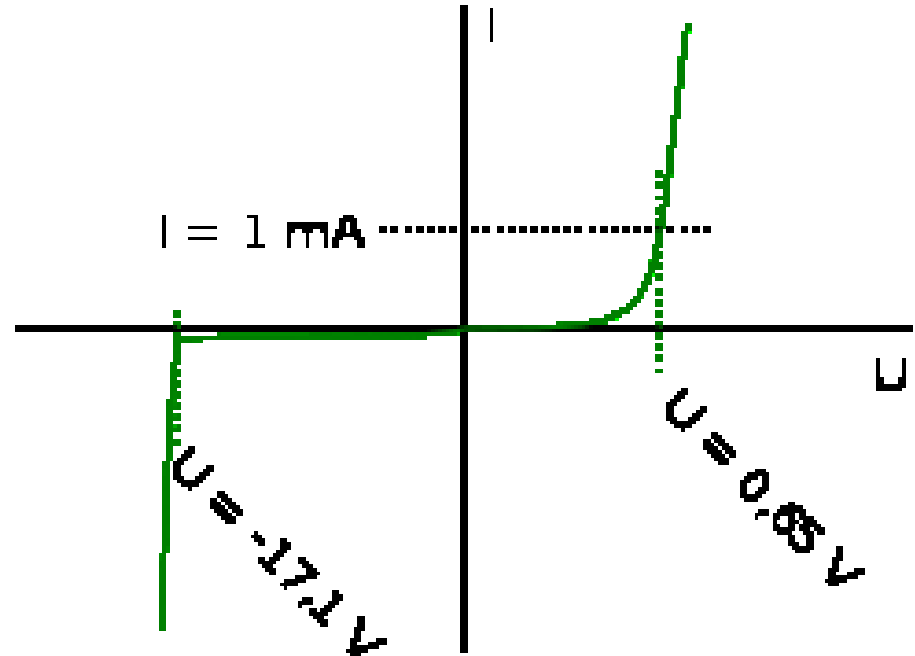


A diode is a nonlinear device and typical linear circuit analysis methods do not apply!

Zener Diode - Voltage Regulator (reverse biased)



A simple regulator circuit that provides a nearly constant output voltage v_o from a variable supply voltage.



A Zener diode is a type of diode that permits current not only in the forward direction like a normal diode, but also in the reverse direction if the voltage is larger than the breakdown voltage known as "Zener knee voltage" or "Zener voltage".

